

REFERENCES

No	Document	Description
1	5.2.2.1 - Site Specific Information_Rev 1	Owner Growth Project - Process Plant - Site Specific Information
2	4.7.1 - Consultation Report 7555 - Owner PDC Data Summary Rev 0, March 2022	Table 2-5, "Back Calculated Ore Parameters", 2014 - 2021 value.
4	4.6.1 - Owner Growth Mass Balance_Nominal Case Report_Rev C	Mass Balance - SysCAD Report Nominal Case. 22/03/2022
5	Owner Comminution Plant Data_B.xlsx"	Benchmarking from FIM pebble crushing circuit.
7	ECI Tender from Concep	Q12375 - EPC Gravity Circuit Rev1
8	5.1.12f - BOD Alignment Template - Flotation Tail PLT, FT CIL, Elution & EW	Basis of Design for Flotation Tailings CIL, Elution & EW circuit. 06/04/2022
9	4.7.3 - Consultation Report 7555 Owner PDC PSD Addendum Rev 1	Owner - Process Design Criteria Summary - PSD Addendum Report No. 7555 Rev 1 March 2022
10	5.1.12g - BOD Alignment Template - Reagents and Services	5.1.12g - BOD Alignment Template - Reagents and Services - Rev C Owner confirmation that regrind mill ball size will be 25 mm (cell F15).
11	EPC Calculation CAL-PR-002	Flotation Tailings CIL Simulation @ 8 and 10 g/L Carbon Concentration
14	Owner Growth Project ECI Outcome Letter - EPC	Owner Growth Project - Process Plant - Formal Agreement for Engineering, Procurement and Construction: Early Contractor Involvement (Value Engineering Phase). 03/08/2022
15	5.1.12b - BOD Alignment Template - 13 Grinding - Rev C	5.1.12b - BOD Alignment Template -13 Grinding - Rev C - EPC (220609)_NSR Review
16	Owner Process Design Criteria - VEP Phase (Stage 1)_I	Owner - Growth Project - Process Design Criteria Rev I. 20/09/2022
18	Cleaner Option 2- Owner Case 2 Scav Con to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner
19	Cleaner Scavenger - Owner Case 2 Scav to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner Scavenger
23	Owner GP Process Design Criteria - Reliance Information S1 Rev K, 25/01/2023	Principal supplied reliance information.
24	VEP Outcome Letters - EPC 09/02/23 and 020523	Principal supplied Scope of Works letter for Stage 1 and 2
25	Consultation Report 7653 - Owner Value Engineering Update	Grinding circuit modelling for Owner by Orway Mineral Consultants, 160223
26	Process Design Workshop Meeting Minutes 170223, MOM-PM-SFP002	Workshop between the Client and EPC
27	Glencore Jameson Cell Sizings and Quote	Simulations from Vendor, 22/02/24 Owner Expansion Project - Jameson Cell Quote OTJ-2220 7532.3
28	Weir Cyclone Simulations and Datasheet - Primary & Gravity Cyclones	Various simulations for the different process cases, and datasheet: 2000-F00-DSH-ME-11005A_B Primary Cyclone
31	Trash Screen Spray Water Information from Vendor	Emails with Vendor, 16/08/23
32	Flotation Mass Balance Scenarios	Mass Balances for Process Margins Rev H and New Mass Balance Cases Rev G
33	Cadia Gravity Electrowinning Cell	Emails with Vendor, 14/03/23. Ag and Cu Assays from site, provided by Owner (23/03/23)
35	Discussions with Client and TQ regarding Cyanide Destruction	Emails with Client, 230323 and TQ: 2000-F72-TQY-ME-10094_0_TQ-R
36	Vendor Information from Transmin for New Flocculant Mixing Plant	Transmin Proposal Execution Phase rev 3, P33-904-06
37	Direction from EPC Mechanical Engineers re: COS	Email, 280323
38	Client Information regarding Grinding Media	Emails with Client, 070323
39	Vendor Information from Concep for the Acacia CS8000 ILR	Emails with Client, 210322
40	EPC Calculation	"SAG_Ball_Screen_Trommel_Calc_Rev_A.xlsx"
41	Client Direction, Transfer of FLS Equipment Technical Information	Emails from client, various dates (May 2023)
42	Primary Crusher Datasheet	Scope of Work - Primary Crusher, Appendix A, doc 2000-F12-SOW-ME-20001 rev 1
43	SoW for Separable Portions Stage 1&2	Scope of Works between Owner and EPC 2000-F00-SOW-CO-20001 & 2000-F00-SOW-CO-20002_0
44	Outstanding Process Inputs from Owner	Outstanding Process Inputs 210623 (version 1) Additional (003) (002), and 2000 F00-TQY-PR-10051.0.TQ-R.0.01
47	Oxygen Consumption Information	Emails from client, 8/5/23
48	Cadia CIL4 Electrowinning Cells	Vendor Quotation 27/03/2024
49	Owner email Owner GP - Float Cell Motors", dated 11 May 2023	FLSmidth flotation blower details.
50	Calculation for LPG Consumption and TPE ECI Tender	CAL-PR-005_A LPG Consumption TPE5000 ELUTION HEATER PACKAGE - ME16 TENDER - COMPLETE 20/05/22
51	FLSmidth - Primary Crusher Application Model	FLS_13x19 TSUV_Calc_FIM_Run 15_13x19 TSUV_ 6.5 OSS_3716 mtph_70, AVG results
52	Vendor Information Mercury Abatement System (CIL4 Carbon Regen Kiln) and Client Direction Variation 25; Vendor information for Lochhead CIL4 Carbon Regen Kiln; Regen Kiln Off Gas Calculations	Information from the Mercury Abatement Vendor, Jan 2024 P3-EPC-COR-161 Variation Direction No.025 Carbon regeneration kiln off-gas mercury abatement system Lochhead Carbon Reactivation Kiln (ME-17) Quotation No. 5211 (August 26, 2022) Kiln Off Gas Calc
53	EPC email Apron Feeder Datasheet Internal Review", dated 30 August 2023	Agreement with Owner to change contract SOW design factor from 120% to 110% of nominal tonnage rate.
54	Weir UFG Cyclone Cluster Modelling	UFG_CY_Select_Design_RevC UFG_CY_Select_Nominal_RevC
55	Takraf Tender Rev 0 Datasheet	2000-F00-DSH-ME-11018_B Trash Screens Completed
56	Owner GP Process Design Criteria - Reliance Information S1 Rev L, 25/01/2023	Principal supplied reliance information.
57	Owner Growth Project SysCAD Model (Historical)	SysCAD Model files and Reports (2022.08.31)-01.spf
58	NTF Deslime & UFG Cyclone PDC Inputs	Word document 260723
59	Joest Vibrating Screens Offer	JOEST_M22-11102 Rev D_JOEST FIRM Offer 2000-F00-DSH-ME-11011_C Vibrating Screens
60	Flotation Launder Wash Water Technical Query	Technical Query (2000-F16-TQY-ME-10007) FLS Launder Water Requirements Rev.0 31/07/2023
62	Technical Query - Design Bulk Density Range for Apron Feeders And Conveyors	2000-F00-TQY-ME-10034

REFERENCES

No	Document	Description
63	Flotation Area - Design P80 Sizes	Email Correspondence. RE - Flotation D80 Mech Requests. 250823
64	Reagent Delivery Formats, 2000-F72-TQY-ME-20001_2	Reagent delivery information, provided by Owner
65	IsaMill Quote and Modelling by Glencore	IsaMill Quote OMT232007747 Owner - Owner EPC - IsaMill Sizing - Design -19102023 - New data supplied water temp 25 Owner - Owner EPC - IsaMill Sizing - Nominal -19102023 - New data supplied water temp 25 Email: "FW_ ME039 IsaMill Latest MB - Clarifications 64 - UFG Mill Media Hopper Live Volume"
66	Regrind Removal TQ	2000-F00-TQY-GE-10080_0 - RESPONSE.pdf
67	Client Comments Stage 1 Rev K PDC and Rev L SysCAD	2000-F00-DCR-PR-10001.K.ICR-R.K.02
68	Raw Water TQ 10044	2000-F24-TQY-PR-10044.0.TQ-R.0.0164
69	SysCAD & PDC Client Comments TQ 10107 Response	2000-F00-TQY-PR-10107.0.TQ-R.0.02
70	Gravity Variation (VAR-017) and Communication from the Client	P3-EPC-COR-085 Variation Direction No.017 Gravity Circuit Modifications Emails from Client/Outcome of meetings
71	Emails from Concep (Knelson & ILR) in relation to Gravity Variation (VAR-017)	Email dated 30/10/23, 'RE: Knelson Concentrators (KC-QS70) and CS10000'
72	Gravity Variation TQ 10115 Response from Client	2000-F14-TQY-PR-10115.0.TQ-R.0.01 2000-F14-TQY-PR-10115_1_TQ-R
73	Quotation and emails from Cadia Systems (EW cells) in relation to Gravity Variation (VAR-017)	Q231115-PR1_EPC_EWC 151123 issue RE_ Owner - Gravity Pregnant Solution Electrowinning - various emails
74	Variation 018 UFG Mill 2 Trash Screen	P3-EPC-COR-104 Variation Direction No.018 UFG Mill 2 Trash Screen
75	Clarification Emails for Flotation Reagent Calculations	Emails with Client 21/11/23: 'RE: Owner GP - PAX Usage/Flow Consumption/Storage Tank RT'
76	Dump Pocket TQ from Owner	2000-F12-TQY-GE-10127_0_TQ-R
77	Wet Static Screens Datasheet (ME40) and Spray Water Information	2000-F00-DSH-ME-11040_1 and emails dated 23/01/24
78	Owner Owner's Manual CAT793F Owner	Owner's Manual CAT793F Owner
79	Water Services PFD Client Review	2000-F00-PFD-PR-10046.D.IFD-R.D.02
80	Area 30 M1 Design Review Report	2000-F30-REP-GE-10101_A - M1 Design Review Report
81	TPE CIL4 Elution Heater Vendor Package	TPE5000 ELUTION HEATER PACKAGE_DES.pdf rev A 20/05/22
82	Client Information LPG Storage Vessel	Email: RE LPG Tank, 22/05/2023
83	TQ147 Reagent Tank Volumes, Var 26 Cyanide Tanks and Var 192 CuSO4 Tanks	2000-F72-TQY-ME-10147_0_TQ-R P3-EPC-COR-162 Variation Direction No.026 Cyanide Tank Storage Capacity P3-EPC-COR-192 Variation Direction No.030 Copper Sulphate Storage
84	Joest Tender Documents	JOEST_M22-11102 Rev H_JOEST FIRM Offer, 06/05/24 RE Growth Project - Stage 2 Vibrating Screen Changes
85	Gekko Carbon Scout ECI Tender Documents	6454-EPC-Carbon Scout-220426, 22/04/22
86	ME65 Carbon Transfer Pump Vendor Information	Weir Budget Quote WMFN20220527
87	ME07 Intertank Screens Vendor Information	P28337_Northern Star_EPC Group Growth (Aus) REV 2
88	TQ206 Request for LPG & RTO Information	2000-F00-TQY-PR-10206.0.TQ-R.0.0270
89	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx
90	Client Comments Stage 1 Rev 0 PDC	2000-F00-DCR-PR-10001.0.IFD-R.0.02
91	Stage 2 Oxygen Consumption Calculations	CAL-PR-003 S2 Oxygen Consumption Calculations
92	SAG & BM Trommel Sizes	2000-ME001.2-B02-50041_C SAG Mill Trommel Sizing Report 2000-ME001.2-M03-50001_B SAG Mill Trommel Assembly 2000-ME001.3-B02-50020_B Ball Mill Trommel Sizing Report 2000-ME001.3-M03-50001_B Ball Mill Trommel Assembly
100	5.1.1 - ECS Process Design Criteria	EngineerProcess Design Criteria - Owner Growth Project ECS. 25/01/2022 J6027-000-DCR-PR-001 Rev D
101	5.1.2 - ECS Mass Balance	EngineerMass Balance - Owner Growth Project ECS. 25/01/2022 J6027-000-CAL-PR-001 Rev D
102	4.5.1 - Process Design Criteria Rev F	Owner - Growth Project - Process Design Criteria Rev F. 16/03/2022
103	4.5.1 - Owner Process Design Criteria - VEP Phase _H, 19/08/2022	Principal supplied reliance information.
104	Email "Carbon Safety Screen" dated 16/02/2024	Confirmation from Owner that currently proposed screen sizes for Carbon Safety Screens is adequate
105	2000-F30-TQY-PR-15001.0.TQ-R.0.02	Client Response to Technical Query - CIL2/3 - Carbon Transfer Rate Between Adsorption Tanks
106	2000-F00-DCR-PR-10002.C.ICR.-R.C.02	Client comments to Process Design Criteria - Stage 2
107	2000-F72-TQY-ME-10260_0-R	Clients Response to Technical Query - Milk of Lime Tank Volume & Residence Time
108	2000-F21-TQY-PR-15014.0.TQ-R.0.02	Clients Response to Technical Query - CIL2/3 & CIL4 Target Process Data
109	2000-F00-TQY-PR-15020.0.TQ-R.0.02	Clients Response to Technical Query - CIL4 Carbon Advance
110	2000-F22-TQY-PR-15018_1_RSP	Clients Response to Technical Query - HCI Dilution Tank
111	Hydrogen Peroxide Technical Data Sheet	Solvay Hydrogen Peroxide Technical Datasheet

REFERENCES

No	Document	Description
112	PAX, CuSO4 & Lead Nitrate Delivery Sizes	Email from IXOM (vendor), dated 28/03/2024
113	Concentrate Thickener Vendor GA	2000-ME091-B02-10001_C
114	Client Comments Stage 2 Rev 0 PDC	20000-F00-DCR-PR-10002_0_AAN
115	2000-F30-TQY-PR-15029_0_TQ-R	Clients Response to Technical Query - Concentrate Pre-Leach Thickener Underflow
116	2000-F72-TQY-PR-15033_0_TQ-R	Clients Response to Technical Query - Lime System Process Design
117	QUO-07322-P7S5G0	Derrick quotation for Derrick Linear Motion Screening Machine 15/08/2024
118	BHT Lime Staking Package	2000-F00-DSH-ME-12013_0 2000-MEP-ME013-REG_A 706-PC-01 706-PC-02 FW_Lime Mill Media Consumption.msg
119	2000-F30-TQY-PR-15097_1	CIL3 Carbon Movement TQ

AREA	PARAMETER	UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
1.0	PRODUCTION SCHEDULE											
1.1	Throughput											
	Annual Throughput	t/a	10,500,000	25,500,000	23,997,254	25,500,000	28,350,000	1	Owner	16, 24, 26 & 32		A
	Annual Operating Days	d/a	365	365	365	365	365	1	Owner	16		A
1.2	Operating Schedules											
	Crushing											
	Plant Availability	%	65.0	65.0	65.0	65.0	65.0	1	Owner	1, 30		A
	Annual Operating Hours	h/a	5,694	5,694	5,694	5,694	5,694	6	Calculated			
	Hourly Throughput - Existing Circuit	th	1,844	1,844	1,844	1,844	1,844	6	Calculated			
	Hourly Throughput - New Circuit	th		2,634	2,370	2,634	3,135	6	Calculated			
	Total Hourly Throughput	th	1,844	4,478	4,214	4,478	4,979	6	Calculated			
1.3	Concentrator											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours - Existing SAG Circuit		8,000	-	-	-	-	6	Calculated			
	Annual Operating Hours - New SAG/Ball Mill Circuit	h/a	-	8,182	8,182	8,182	8,182	6	Calculated			
	Hourly Throughput - Existing SAG Circuit	th	1,313	1,313	1,313	1,313	1,313	6	Calculated			
	Hourly Throughput - New SAG/Ball Mill Circuit	th		1,833	1,650	1,833	2,182	6	Calculated			
	Total Hourly Throughput	th	1,313	3,146	2,962	3,146	3,494	6	Calculated			
	UFG Circuit											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours	h/a	8,000	8,182	8,182	8,182	8,182	6	Calculated			
2.0	MATERIAL CHARACTERISTICS											
2.1	General Ore Properties											
	Moisture Content	%w/w			1.3			1	Owner	16		A
	Ore Specific Gravity	Nominal			2.90			1	Owner	16		A
		Range			2.70 - 3.10			1	Owner	23		B
	Ore Bulk Density	Nominal			1.90			1	Owner	16		A
		Range			1.70 - 2.10			1	Owner	56		B
		Design Range									To be used for all conveyors & apron feeders in the crushing and milling circuits	B
	Angle of Repose	degrees			1.60 - 1.90			1	Owner	62		A
	Drawdown Angle	degrees			36			1	Owner	16		A
					60			1	Owner	16		A
2.2	Ore Comminution Parameters											
	Crushing Work Index											
	Testwork Average	kWh/t			17.8			1	Owner	16		A
	Testwork Range	kWh/t			16 - 30			1	Owner	16		A
	Testwork 80th Percentile	kWh/t			18.8			1	Owner	2, 23	2014 - 2021 value (as of rev E)	A
	Abrasion Index											
	Testwork Average				0.145			1	Owner	16		A
	Testwork Range				0.020 - 0.680			1	Owner	16		A
	Testwork 80th Percentile				0.192			1	Owner	16		A
	Ball Mill Work Index	Range	kWh/t		11.5 - 15			1	Owner	41		B
	A*b	Design	kWh/t		12.5			1	Owner	2	2014 - 2021 value	A
		Range			38 - 48			1	Owner	41		B
		Design			43.8			1	Owner	2	2014 - 2021 value	A
2.3	Pebble Comminution Parameters											
	Crushing Work Index	Design	kWh/t		18.8			1	Owner	16		A
		Maximum	kWh/t		23.0			1	Owner	23		A
	Bulk Density	Minimum	tm ³		1.60			1	Owner	16, 62		A
		Maximum	tm ³		1.90			1	Owner	41, 62		B
2.4	Non Ore Grades											
	Feed Grade - S	% S	1.02	1.02		1.40	1.02	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Feed Grade - Au	Au (g/t)	1.58	1.58		1.70	1.58	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Final Tail Grade - S	%	0.18	1.01	1.39		1.01	1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au	Au (g/t)	0.23	0.23	0.25		0.23	1	Owner	23	Blue data is from Owner Ref 23	A
2.5	Design Ore Grades											
	Feed Grade - S	%S	1.49		1.49			1	Owner	23		A
	Feed Grade - Au	Au (g/t)	1.81		1.81			1	Owner	23		A
	Final Tail Grade - S	%	0.29		1.48			1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au	Au (g/t)	0.38		0.26			1	Owner	23	Blue data is from Owner Ref 23	A
2.6	Recovery											
	Total Gold Recovery	Nominal	%	84	84	84	84	6	Calculated			
	Gravity Gold Recovery	Nominal	%	14.00	14.00	14.00	14.00	1	Owner	23	Relative to feed grade	A
	Gravity Gold Recovery	Maximum	%	20.00	20.00	20.00	20.00	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Nominal	%	71.80	71.80	71.80	71.80	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Maximum	%	75.00	75.00	75.00	75.00	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Nominal	%	85.10	85.10	85.10	85.10	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Maximum	%	90.00	90.00	90.00	90.00	1	Owner	23	Relative to feed grade	A
	Flotation Concentrate Leach Gold Recovery	Nominal	%	84.00	84.00	84.00	84.00	1	Owner	44	Range 80-88%	A
	Flotation Concentrate Leach Gold Recovery	Maximum	%	88.00	88.00	88.00	88.00	1	Owner	44	Range 80-88%	A
	Flotation Tails Leach Gold Recovery	Nominal	%	34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
	Flotation Tails Leach Gold Recovery	Maximum	%	34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
2.7	Gold Production											
	Nominal	oz Au/a		1,108,133	1,194,627	1,195,298	1,231,759	6	Calculated			
2.8	Flotation Concentrate Production	t/a		636,932	675,971	676,099	707,270	6	Calculated			
2.9	Process Water SG			1.070		1.070	1.070	1	Owner	1		A
2.10	Scheme Water SG			1.000		1.000	1.000	1	Owner	1		B
3.0	CRUSHING CIRCUIT											
3.1	Circuit Parameters											
	Throughput	Nominal	th	1,844	2,634	2,370	2,634	6	Calculated		New circuit	B
		Design	th		3,135	3,135	3,135	1	Owner	43		
		Maximum	th		3,436	3,436	3,436	1	Owner	56	Crusher fall through from FLS and not the circuit design basis	B
	Feed PSD											
	F ₁₀	mm		1,000	1,000	1,000	1,000	1	Owner	23		A
	F ₅₀	mm		450	450	450	450	1	Owner	23		A
	F ₈₀	mm		150	150	150	150	1	Owner	23		A
	F ₂₀₀	mm		25	25	25	25	1	Owner	23		A
	Product PSD											
	P ₈₀ - Maximum	mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		A
3.2	Dump Pocket											
	Dump Pocket Type			Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	1	Owner	16		A
	Truck Model			793F	793F	793F	793F	1	Owner	23		A
	Truck Payload	t		250	250	250	250	1	Owner	78	Owner's Manual CAT793F.	B
	SAE 2:1 Heaped Tray Capacity	m ³		179	179	179	179	1	Owner	76, 78	Owner's Manual CAT793F.	B
	Dump Pocket Capacity	m ³		179	179	179	179	1	Owner	76, 78	Live capacity from one side	B
3.3	Primary Crusher											
	Type		Gyratory	Gyratory	Gyratory	Gyratory	Gyratory	1	Owner	16	Owner Supplied	A
	Model		54-75Mk II	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1	Owner	41	Owner Supplied	A
	Open Side Setting	mm	141	165	165	165	165	5	Vendor	51		B
	Throw	mm	38	38	38	38	38	1	Owner	42		B
	Motor Power	kW	450	600	600	600	600	5	Vendor	42		A
	Simulation Results											
	P ₅₀	Max	mm	122	122	122	122	5	Vendor	51	Simulated, see reference document for inputs	B
	P ₁₀₀	Max	mm	229	229	229	229	5	Vendor	51	Simulated, see reference document for inputs	B

AREA	PARAMETER	UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
			1	2	3	4							
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
3.4	Crushed Ore Storage Type		Stockpile	Stockpile	Stockpile	Stockpile	Stockpile	1	Owner	16		A	
	Live Capacity - Existing	t	21,001	-	-	-	-	6	Calculated			1	
	Residence Time - Existing	h	16	-	-	-	-	6	Calculated				
	Bulk Density	tm ³		1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1	Owner	43		B	
	Live Capacity - New	m ³		12,035	12,035	12,035	12,035	2	EPC	37		A	
	Residence Time - Minimum/Range	h		11.16	12.40	11.16	9.38	6	Calculated				
	Live Capacity at 1.70 tm ³	t		20,460	20,460	20,460	20,460	6	Calculated				
	Live Capacity at 1.90 tm ³	t		22,867	22,867	22,867	22,867	6	Calculated				
	Live Capacity - New	m ³		11,293	11,293	11,293	11,293	2	EPC	37		A	
	Residence Time - Minimum/Range	h		10.47	11.64	10.47	8.80	6	Calculated				
	Live Capacity at 1.70 tm ³	t		19,198	19,198	19,198	19,198	6	Calculated				
	Live Capacity at 1.90 tm ³	t		21,457	21,457	21,457	21,457	6	Calculated				
	Primary Crusher Discharge Feeder Number			1	1	1	1	1	Owner	43		B	
	Type			Apron	Apron	Apron	Apron	1	Owner	16		B	
	Capacity	Nominal		2,634	2,634	2,634	3,135	6	Calculated				
		Design		3,135	3,135	3,135	3,135	1	Owner	43		B	
3.5	Reclaim Feeders Number		4	2	2	2	2	1, 2	EPC	Owner	16, 26		A
	Type		Apron	Apron	Apron	Apron	Apron	1	Owner	16		A	
	Capacity per feeder	Nominal		917	917	917	1,091	6	Calculated				
		Design		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.		B
3.6	Emergency Feeders Number		1	1	1	1	1	1	Owner	16		A	
	Type		Apron	Apron	Apron	Apron	Apron	1	Owner	16		A	
	Capacity per feeder	Nominal		1,833	1,833	1,833	2,182	6	Calculated				
		Design		2,017	2,017	2,017	2,017	9	Agreed	53	110% of nominal tonnage rate.		B
	Loader Model			992K/994K	992K/994K	992K/994K	992K/994K	1	Owner	23			A
4.0	GRINDING AND CLASSIFICATION												
4.1	Circuit Parameters Type		SABC	SABC	SABC	SABC	SABC	1	Owner	16		A	
	Nom Throughput	Nominal @ 180 µm	1,313	1,833	1,650	1,833		6	Calculated				
	Nom Throughput	Nominal @ 220 µm					2,182	6	Calculated				
	Bulk Density	Design Range		1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1	Owner	43, 62		B	
	Milling Circuit PSD							1, 5	Owner,	Vendor	23, 25		A
	P ₈₀	Target	100	130	130	130	145	1	Owner		16, 23, 24		A
	P ₁₀₀		180	180	180	180	220						
	SAG Mill Specific Energy - 50th Percentile		8.3					1	Owner	16			A
	SAG Mill Specific Energy - 80th Percentile		8.9					1	Owner	16			A
	Ball Mill Specific Energy - 50th Percentile		5.4					1	Owner	16			A
	Ball Mill Specific Energy - 80th Percentile		6.1					1	Owner	16			A
	P ₅₀	Modelled		189	189	189	220	1	Owner	25			A
	SAG Mill Specific Energy Modelled		9.14	9.14	9.14	9.14	7.68	1	Owner	25			A
	Ball Mill Specific Energy Modelled	@ 180 µm	9.68	9.68	9.68	9.68		1	Owner	25			A
	Ball Mill Specific Energy Modelled	@ 220 µm					8.14	2	EPC	25			A
4.2	Primary Mill Type		SAG Mill	SAG Mill	SAG Mill	SAG Mill	SAG Mill	1	Owner	16	Owner Supplied		A
	Mill Diameter	m	10.97	10.97	10.97	10.97		1	Owner	16	Owner Supplied		A
	Effective Grinding Length - EGL	m	4.27	8.38	8.38	8.38		1	Owner	16, 23	Owner Supplied		A
	Duty Mill Speed Range	% Cs	60 - 80	60 - 83	60 - 83	60 - 83	60 - 83	1	Owner	24			A
	Duty Ball Charge	% v/v	13 - 17	10.5	10.5	10.5	10.9 - 11.3	1	Owner	16, 24			A
	Duty Total Load	% v/v	25 - 30	27	27	27	25	1	Owner	16, 24			A
	Total Ball Charge	Structural		18	18	18		1	Owner	23		For structural design	A
	Total Load	Structural		35	35	35		1	Owner	23		For structural design	A
	Mill Discharge Density	% w/w	75	75	75	75		1	Owner	9, 69			A
	Mill Pinion Power	Average	10,894	-	-	-		1	Owner	25			A
	Mill Pinion Power	Duty	11,682	16,078	15,078	16,757		6	Calculated				1
	Installed Motor Power		12,000	20,000	20,000	20,000	20,000	1	Owner	41, 23			A
	Liner Type			Steel	Steel	Steel	Steel	1	Owner	41			A
	Curved Pulp Lifter Capacity	Max		2,100	2,100	2,100	2,100	1	Owner	41			A
	Pebble Port Aperture		75	75	75	75		1	Owner	41		FLS offer	A
	Discharge Grate Open Area			11 - 12	11 - 12	11 - 12	11 - 12	1	Owner	41			B
	Trommel Dimensions	ID x L		3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	5	Vendor	92			1
	Trommel Open Area	%	25	25	25	25		1	Owner	41			B
	Trommel Aperture	mm		12 x 40	12 x 40	12 x 40	12 x 40	1	Owner	41			B
	Spray Water	m ³ /h	150	150	150	150		2	EPC	40			B
	Trommel Undersize	t/h		1,050	1,050	1,050	1,050	2	EPC	32			B
	Existing Fim SAG Mill Transfer Density	% w/w	60					1	Owner	57, 69			B
4.3	Primary Mill Discharge Screen Discharge Screen Type			Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1	Owner	16			A
	Screen Size	W x L		4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	5	Vendor	59			A
	Effective Screen Area	m ²		36.47	36.47	36.47	36.47	5	Vendor	59			B
	Screen Open Area	%		34.0	34.0	34.0	34.0	5	Vendor	59			B
	Screen Aperture	mm		12 x 55	12 x 55	12 x 55	12 x 55	2	EPC	40			A
	Specific Screen Flux Rate	m ³ /m ² /h OA		18.68	18.68	18.68	29.90	5	Vendor	59		Includes wash water	B
	Oversize Moisture	%w/w	5	13	13	13		5	Vendor	59			B
	Spray/Wash Water	m ³ /h		197.2	197.2	197.2	197.2	5	Vendor	59			B
	Discharge Screen Feed % Solids	%w/w		85	85	85		2	EPC	32			B
	Discharge Screen Oversize	t/h		375	375	375	375	1	Owner	41			B
4.4	Secondary Mill Type			Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	1	Owner	41	Owner Supplied		A
	Mill Diameter	m		8.53	8.53	8.53	8.53	1	Owner	23	Owner Supplied		A
	Effective Grinding Length - EGL	m		13.26	13.26	13.26	13.26	1	Owner	23	Owner Supplied		A
	Duty Mill Speed	% Cs		60 - 78	60 - 78	60 - 78	60 - 78	1	Owner	41			A
	Duty Ball Charge	Nominal		28.5	28.5	28.5	28.5	1	Owner	41			A
	Duty Ball Charge	Maximum		31.0	31.0	31.0	31.0	1	Owner	24			A
	Total Ball Charge	Structural		38	38	38	38	1	Owner	23		With polymet liners. For structural design	A
	Mill Discharge Density	% w/w		75	75	75		1	Owner	4			A
	Mill Pinion Power	Duty		17,747	15,969	17,747	17,759	6	Calculated				
	Installed Motor Power			20,000	20,000	20,000	20,000	1	Owner	23			A
	Liner Type			Poly-Met	Poly-Met	Poly-Met	Poly-Met	1	Owner	23			A
	Trommel Flux Rate	Nominal		230	230	230	230	6	Calculated				
	Trommel Dimensions	ID x L		4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	5	Vendor	92	ID = internal diameter		1
	Trommel Capacity	Maximum		6,681	6,681	6,681	6,681	1	Owner	23	Includes spray water		A
	Mill Discharge Flowrate	Nominal		6,373	6,197	6,383	6,277	6	Calculated		Includes spray water		
	Trommel Open Area	Minimum		30	30	30		1	Owner	23			A
	Trommel Aperture	mm		19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	1	Owner	23			A
	Trommel Screen Spray Water	m ³ /h		150	150	150		1	Owner	4			A
4.5	Pebble Crusher Type		Cone	Cone	Cone	Cone	Cone	1	Owner	16	Owner Supplied		A
	Number		2 (duty / standby)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	1	Owner	16, 23	Owner Supplied		A
	Model		Symons 78	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	1	Owner	41	Owner Supplied		A
	Installed Motor Power per Crusher	kW	375 / 260	375	375	375	375	1	Owner	16, 41, 44			A
	Closed Side Setting	mm	7 - 9	7	7	7		1	Owner	16, 41			B
	Pebble Crushing Work Index	Maximum		23	23	23	23	1	Owner	56			1
	Pebble Bulk Density	Range		1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1	Owner	56			1
	Throughput	Nominal		302				6	Calculated				
	Throughput	Nominal		302				1	Owner				
	Range	t/h		305	305	305	610	1	Owner	41			B
	% of new feed	% of new feed		23 - 27				1	Owner	16			B
	% of new feed	% of new feed		23.0				1	Owner	16			B
	Crusher Capacity at CSS	Maximum		16.6	18.5	16.6	28.0	6	Calculated			305 t/h limit per crusher unit at CSS. This throughput represents the increased pebble load as the pebble ports wear out on SAG mill discharge grate	A
	Feed Bin Capacity	per bin		610	610	610	610	1	Owner	41			
	Feed Bin Capacity	min	50	12.8	12.8	12.8	6	1, 2	Owner,	EPC	26		A
	Feed Bin Capacity	t	260	65	65	65	65	6	Calculated			Total Bin capacity of 130 t (2 bins)	
	Feed Moisture	Range	132	34	34	34		6	Calculated			Total Bin capacity of 68 m ³ (2 bins)	
	Feed Pebble PSD			2 - 8	2 - 8	2 - 8	2 - 8	1	Owner	41			B
	F ₁₀₀	mm		95.0	95.0	95.0	95.0	1	Owner	41		Linear interpolation from PSD data	B
	F ₅₀	mm		49.0	49.0	49.0	49.0	1	Owner	41		Linear interpolation from PSD data	B
	F ₂₀	mm		37.0	37.0	37.0	37.0	1	Owner	41		Linear interpolation from PSD data	B
	Product Pebble PSD												
	P ₅₀	mm		12.0	12.0	12.0	12.0	1	Owner	69	FLS guaranteed value		B
4.6	Classification Feed Size F ₅₀	Typical		276	276	276	276	1	Owner	16			A
	Feed Size F ₁₀₀	Modelled		350	350	350	350	1	Owner	9	Consult	Modelling	B
	Feed Size F ₅₀	Modelled		0.757	0.757	0.757	0.757	1	Owner	9	Consult	Modelling	B
	Feed Size F ₁₀₀	Modelled		0.306	0.306	0.306	0.306	1	Owner	9	Consult	Modelling	B
	Product Size P ₁₀₀	Target		180	180	180	220	1	Owner	26			A
	Product Size P ₅₀	Maximum		250	250	250	250	1	Owner	26			A
	Type			Hydrocyclone	Hydrocyclone	Hydrocyclone	Hydrocyclone	1	Owner				A
	Cyclone Model			800CVX13	800CVX13	800CVX13	800CVX13	5	Vendor	28			A
	Number of Cyclone Clusters			1	1	1	1	5	Vendor	28			A
	Distributor Outlets per Cluster	No.		18	18	18	18	5</					

AREA	PARAMETER		UNIT		DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
					1	2	3	4						
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
4.7	Number Operating Cyclones - Gravity	Nominal	No.		2	2	2	2	5	Vendor	28	Gravity cyclones require larger spigot and VF to achieve mass flow of 1000tph to underflow. There is also a higher recirc. load.	A	
		Maximum	No.		2	2	2	2	5	Vendor	28		A	
	Cyclone Feed	Nominal	th		14,421	13,986	14,421	14,421	6	Calculated				
	Cyclone Feed Flow	Nominal	m³/h		13,233	12,816	13,217	13,217	6	Calculated				
	Cyclone Feed Flow	Maximum @ 61.6%	m³/h		13,862	13,862	13,862	13,862	1	Owner	26		Flow rate of 13,862m³/hr requires 14 or 15 online. 2 gravity as advised by Weir	A
	Cyclone Feed Density		% w/w		62.0	62.0	62.0	65.0	5	Vendor	28		A	
	Underflow Solids Concentration	Minimum	% w/w		75.0	75.0	75.0	77.2	1	Owner	16		B	
	Underflow Solids Concentration		% w/w		77.2	77.0	77.2	77.2	6	Calculated				
	Underflow Solids		th		11,275	11,024	11,275	11,129	6	Calculated				
	Overflow Solids Concentration		% w/w		36.4	35.9	36.4	43.3	5	Vendor	28		A	
	Overflow Solids		th		3,146	2,962	3,146	3,494	6	Calculated				
	Circulating Load - Cyclone UF / Cyclone OF		%		358	372	358	318	5	Vendor	28	A		
	Circulating Load - Ball Mill Discharge / Cyclone OF		%		301	311	301	267	6	Calculated				
	Operating Pressure - Cyclones in Cluster		kPa		82	91	82	73	5	Vendor	28	B		
	Operating Pressure - Gravity Cyclones		kPa		83	86	83	70	5	Vendor	28	B		
	Trash Screen													
	Type				Linear Screen	Linear Screen	Linear Screen	Linear Screen	1	Owner	16		A	
	Total Number Screens				4	4	4	4	6	Calculated				
	Number Operating				3	3	3	3	1	Owner	14		A	
	Number Standby				1	1	1	1	1	Owner	14		A	
	Screen Dimensions	W x L	m		9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	5	Vendor	55		B	
	Screen Area per Screen	Effective	m²		40.03	40.03	40.03	40.03	5	Vendor	55		A	
	Total Screen Area		m²		120.09	120.09	120.09	120.09	6	Calculated				
	Effective Screen Area		%		34	34	34	34	5	Vendor	55		B	
	Screen Aperture		mm		1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	5	Vendor	55		B	
	Screen Feed Flow (incl. spray water)		m³/h		2,183	2,093	2,182	1,936	6	Calculated				
	Maximum Screen Flux Rate		m³/m²/h		75	75	75	75	1	Owner	26		A	
	Specific Screen Flux Rate		m³/m²/h		54.5	52.3	54.5	48.4	6	Calculated				
	Spray/Wash Water	per screen	m³/h		60	60	60	60	5	Vendor	31, 55		B	
	Total Spray/Wash Water		m³/h		180	180	180	180	6	Calculated				
	Spray Water reporting to Screen Oversize	per screen	m³/h		45	45	45	45	5	Vendor	31		B	
	Total Spray Water reporting to Screen Oversize	Total	m³/h		135	135	135	135	6	Calculated				
	Spray Water reporting to Screen Undersize	per screen	m³/h		15	15	15	15	5	Vendor	31		B	
	Spray Water reporting to Screen Undersize	Total	m³/h		45	45	45	45	6	Calculated				
	Water to Trash Screens Final Collection Box		m³/h		258.0	134.9	258.0	306.8	6	Calculated		HP Process Water, back calculated in SysCAD	B	
4.8	Trash Dewatering													
	Dewatering Screen Type			Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		A		
	Number			1	1	1	1	1	Owner	16		A		
	Screen Size	W x L	m	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	5	Vendor	59		B		
	Effective Screening Area per Screen		m²	6.14	6.14	6.14	6.14	5	Vendor	59		B		
	Specific Screen Flux Rate	Design	m³/m²/h	50	50	50	50	1	Owner	16		A		
		Nominal	m³/m²/h	22.36	22.36	22.36	22.36	5	Vendor	59		A		
	Screen Aperture		mm	1 x 15	1 x 15	1 x 15	1 x 15	1	Owner	16		A		
	Deck Type			Polyurethane	Polyurethane	Polyurethane	Polyurethane	5	Vendor	59		A		
	Quantity of Wood/Grit/Trash removed per day	td/day		27	27	27	27	1	Owner	16		A		
	Total Spray/Wash Water		m³/h	11	11	11	11	5	Vendor	59		B		
5.0 GRAVITY CONCENTRATION														
5.1 Gravity Scalping Screen														
5.1	Type			Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	5	Vendor	59		A		
	Number of Screens			2	2	2	2	1	Owner	16		A		
	Screen Solids Feed Rate per Screen	Nominal	th	1,000	1,000	1,000	1,000	6	Calculated			Only 2x gravity concentrator circuits, no future allowance for additional circuits		
		Design	th	1,111	1,111	1,111	1,111	6	Calculated			To achieve 1,000 t/h to gravity concentrators at design O/S split		
	Screen Feed Density	Target	% w/w	60.0	60.0	60.0	60.0	1	Owner	70		B		
	Screen Feed Density	Actual	% w/w	60.0	60.0	60.0	60.0	6	Calculated			Excludes Spray Water		
	Screen Feed Flowrate per Screen	Nominal	m³/h	968	966	967	967	6	Calculated			Excludes Spray Water		
		Design	m³/h	1,075	1,073	1,074	1,075	6	Calculated			Excludes Spray Water		
	Screen Size	W x L	m	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	1	Owner	59, 70, 72		B		
	Effective Screen Area per Screen	Overall	m²	22.31	22.31	22.31	22.31	5	Vendor	59				
	Specific Screen Flux Rate	Nominal	m³/m²/h	48.0	47.9	47.9	47.9	6	Calculated			Includes Spray Water		
		Design	m³/m²/h	52.8	52.7	52.7	52.7	6	Calculated			Includes Spray Water		
		Target	m³/m²/h	50	50	50	50	1	Owner	16, 70		A		
	Screen Split to Oversize - Range		%	8 - 15	8 - 15	8 - 15	8 - 15	1	Owner	16		A		
		Design	%	10	10	10	10	1	Owner	16, 72		A		
	Screen Aperture		mm	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	5	Vendor	59		B		
	Screen Spray Water	per screen	m³/h	102	102	102	102	5	Vendor	59		B		
	Oversize Solids Concentration		% w/w	87	87	87	87	5	Vendor	59		B		
	Oversize Slurry Destination			Ball mill feed	Ball mill feed	Ball mill feed	Ball mill feed	1	Owner	43		A		
	Undersize Slurry Destination			Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	1	Owner	43		A		
5.2	Gravity Concentrator													
	Type			Knelson	Knelson	Knelson	Knelson	1	Owner	16		A		
	Number of Concentrators			2	2	2	2	1	Owner	16		A		
	Model			KC-QS70	KC-QS70	KC-QS70	KC-QS70	1	Owner	16		A		
	Feed Capacity per Unit	Max	th	1,000	1,000	1,000	1,000	5	Vendor	39, 71		A		
	Volumetric Capacity per Unit	Max	m³/h	910	910	910	910	5	Vendor	7		B		
	Feed Solids Rate to Circuit		th	1,800	1,800	1,800	1,800	2, 5	EPC Vendor	7, 71		A		
	Feed Solids Rate per Unit	Max	th	2,000	2,000	2,000	2,000	6	Calculated			300-1000t/hr/unit operating range		
	Feed Density	Nominal	% w/w	900	900	900	900	6	Calculated			Vendor recommends 90% of the max (Source 71).		
	Volumetric Flowrate per Unit		m³/h	54.2	54.2	54.2	54.2	6	Calculated					
				1,021	1,020	1,020	1,021	6	Calculated					
Fluidising Water per Unit		m³/h	120	120	120	120	5	Vendor	7		B			
Fluidising Water	Nominal	m³/h	240	240	240	240	6	Calculated						
Cycle Time (Inclusive of Flush Cycle)		min	30	30	30	30	5	Vendor	71		B			
Cycles Per Day		No.	48	48	48	48	6	Calculated						
Concentrate Production Per Cycle		kg	150	150	150	150	5	Vendor	7, 71		A			
Concentrate Bulk SG		t/m³	2.20	2.20	2.20	2.20	1	Owner	44		B			
Daily Concentrate Production		kg/d	14,400	14,400	14,400	14,400	6	Calculated						
		m³/d	6.55	6.55	6.55	6.55	6	Calculated						
Fresh Feed to Concentrators		%	57.2	60.8	57.2	51.5	6	Calculated						
Tailings Destination			Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	1	Owner	43		A			
6.0 INTENSIVE LEACH														
6.1 Intensive Leach Reactor														
6.1	Type			Acacia	Acacia	Acacia	Acacia	1	Owner	16		A		
	Model			CS10000	CS10000	CS10000	CS10000	1	Owner	70		B		
	Number			1	1	1	1	1	Owner	16		A		
	Batches per Day	Nominal		1	1	1	1	5	Vendor	39		A		
		Max		2	2	2	2	5	Vendor	39		A		
	Solids Feed Rate per Batch		kg/batch	14,400	14,400	14,400	14,400	6	Calculated					
	Design Solids Feed Rate per Batch		kg/batch	15,000	15,000	15,000	15,000	5	Vendor	71		B		
	Feed % Solids (after dewatering)		% w/w	90	90	90	90	5	Vendor	71		B		
	Gravity Gold Recovery	Nominal	% Au	14	14	14	14	1	Owner	16, 72		B		
		Design	% Au	20	20	20	20	1	Owner	16, 72		A		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
7.0	FLOTATION										
7.1	General										
	Throughput	t/h		3,146	2,962	3,146	3,494	6	Calculated		
	Feed Particle Size - p80	µm		180	180	180	220	1	Owner	26	A
		Design		220	220	220	220	1	Owner	26	B
	Froth Factors										
	Rougher Flotation Concentrate			2.00	2.00	2.00	2.00	1	Owner	16	A
	Scavenger Flotation Concentrate			1.60	1.60	1.60	1.60	1	Owner	16	A
	Cleaner Flotation Concentrate			2.50	2.50	2.50	2.50	1	Owner	16	A
	Cleaner Scavenger Flotation Concentrate			2.00	2.00	2.00	2.00	1	Owner	16	A
	Flotation Cell Transfer Lines			1.20	1.20	1.20	1.20	1	Owner	16	A
	Launders			-	-	-	-	1	Owner	16	A
	Spray Water Line Pressure (at vendor battery limit)	kPa		62.74	62.74	62.74	62.74	5	Vendor	60	B
7.2	Rougher Flotation										
	Feed Streams			Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	1	Owner	43	Owner Supplied
	Solids Feed Rate	New Feed	t/h	3,146	2,962	3,146	3,494	6	Calculated		A
		Total	t/h	3,161	2,977	3,160	3,512	6	Calculated		
	Volumetric Feed Rate		m³/h	6,719	6,326	6,716	6,027	6	Calculated		Includes sample return
	Feed Solids Concentration		% w/w	34.4	34.4	34.4	40.5	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)
	Mass Pull		%	4.4	4.4	4.4	4.4	6	Calculated		
	Concentrate 1 Upgrade Ratio			15	15	15	15	1	Owner	16	A
	Concentrate 1 Grade		% S	15.3	22.4	21.0	15.3	6	Calculated		
	Concentrate 1 Stage Recovery		% Au in fl. feed	65	65	65	65	1	Owner	23	A
	Concentrate 1 Stage Recovery		% of S in fl. feed	65	65	65	65	1	Owner	23	A
	Concentrate 1 Production		t/h	138	130	138	153	6	Calculated		
	Concentrate 1 Solids Concentration		% w/w	25	25	25	25	1	Owner	16	A
	Concentrate 1 Slurry Volume - Excl Spray Water		m³/h	427	400	425	474	6	Calculated		
	Concentrate 1 SG			3.32	3.54	3.50	3.32	2	EPC	32	A
	Concentrate 1 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
		Design	m³/h	50	50	50	50	5	Vendor	60	A
	Mass Pull		%	2.4	2.4	2.4	2.4	6	Calculated		
	Concentrate 2 & 3 Upgrade Ratio			7	7	7	7	1	Owner	16	A
	Concentrate 2 & 3 Grade		% S	7.1	10.4	9.8	7.1	6	Calculated		
	Concentrate 2 & 3 Stage Recovery		% Au in fl. feed	17	17	17	17	1	Owner	23	A
	Concentrate 2 & 3 Stage Recovery		% of S in fl. feed	17	17	17	17	1	Owner	23	A
	Concentrate 2 & 3 Production		t/h	77	72	77	86	6	Calculated		
	Concentrate 2 & 3 Solids Concentration		% w/w	25	25	25	25	1	Owner	16	A
	Concentrate 2 & 3 Slurry Volume - Excl Spray Water		m³/h	241	225	239	268	6	Calculated		
	Concentrate 2 & 3 SG			3.06	3.17	3.15	3.06	2	EPC	32	A
	Concentrate 2 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	
	Concentrate 3 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	
	Concentrate 2 & 3 Spray Water	Design (Combined)	m³/h	50	50	50	50	5	Vendor	50	A
	Rougher Tail Solids Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		
	Rougher Tail Slurry Volumetric Flow		m³/h	6,051	5,701	6,052	5,285	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Owner	16	A
	Number Cells Installed			3	3	3	3	1	Owner	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Owner	16	A
	Total Cell Volume		m³	1,890	1,890	1,890	1,890	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Owner	41	A
	Plant Residence Time		min	15.1	16.0	15.1	17.0	6	Calculated		Rougher residence time existing plant = 10-12 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Owner	16	A
7.3	Scavenger Flotation										
	Feed Streams			Rougher Tailings	Rougher Tailings	Rougher Tailings	Rougher Tailings	1	Owner	43	Owner Supplied
	Solids Feed Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		A
	Volumetric Feed Rate		m³/h	6,051	5,701	6,052	5,286	6	Calculated		
	Feed Solids Concentration		% w/w	35.3	35.3	35.3	42.4	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)
	Mass Pull (Scavenger Concentrate)		%	1.1	1.1	1.1	1.1	6	Calculated		
	Scavenger Concentrate Upgrade Ratio			5	5	5	5	1	Owner	16	A
	Scavenger Concentrate Grade		% S	5.1	7.5	7.0	5.1	6	Calculated		
	Scavenger Concentrate Stage Recovery		% Au in fl. feed	5	5	5	5	1	Owner	23	A
	Scavenger Concentrate Stage Recovery		% of S in fl. feed	5	5	5	5	1	Owner	23	A
	Scavenger Concentrate Production		t/h	32	30	32	35	6	Calculated		
	Scavenger Concentrate Solids Concentration		% w/w	25	25	25	25	2	EPC	32	A
	Scavenger Concentrate Slurry Volume - Excl Spray Water		m³/h	99	93	99	110	6	Calculated		
	Scavenger Concentrate SG			3.00	3.07	3.06	3.00	2	EPC	32	A
	Scavenger Concentrate Solids Concentration - Diluted		% w/w	20	20	20	20	2	EPC	32	A
	Scavenger Concentrate Launder Spray Water	Actual (Combined)	m³/h	29	28	29	33	6	Calculated		
		Operating Range	m³/h/cell	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
		Design	m³/h/cell	25.00	25.00	25.00	25.00	5	Vendor	60	B
	Scavenger Tail Solids Rate		t/h	2,914	2,745	2,914	3,238	6	Calculated		
	Scavenger Tail Slurry Volumetric Flow		m³/h	5,952	5,608	5,953	5,176	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Owner	16	A
	Number Cells Installed			4	4	4	4	1	Owner	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Owner	16	A
	Total Cell Volume		m³	2,520	2,520	2,520	2,520	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Owner	41	A
	Plant Residence Time		min	21.4	22.7	21.4	24.6	6	Calculated		Scavenger residence time existing plant = 29 - 32 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Combined Rougher and Scavenger Residence Time			36.5	38.8	36.5	41.6	6	Calculated		
7.4	Cleaner Flotation										
	Feed Streams			Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	1	Owner	43	Owner Supplied
	Solids Feed Rate	New Feed Only	t/h	253	244	258	282	6	Calculated		New feed only
		Total	t/h	465	424	439	478	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Volumetric Feed Rate to Cell - Target	Total	m³/h	1,998	1,998	1,998	1,998	5	Vendor	27	Average of four modelled cases
	Volumetric Feed Rate to Cell - Actual	Total	m³/h	1,998	1,998	1,998	1,998	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Feed Solids Concentration - Actual	Total	% w/w	19.0	17.5	18.0	19.5	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Particle Size p80	Design	µm	220	220	220	220	6	Calculated		Feed, concentrate and tailings p80 assumed equal (conservative).
		Calculated	µm	180	180	180	220	6	Calculated		Max of calculated p80
	Mass Pull		%	30.8	44.0	41.5	30.7	6	Calculated		
	Concentrate Grade		% S	35	35	35	35	1	Owner	26	A
	Concentrate Stage Recovery		% S	93.3	93.6	93.4	93.1	6	Calculated		
	Concentrate Recovery	Overall	% of S in fl. feed	85.1	85.1	85.1	85.1	1	Owner	23	A
	Concentrate Recovery	Overall	% Au in fl. feed	84	83	83	84	6	Calculated		
	Concentrate Production		t/h	78	107	107	86	6	Calculated		
	Concentrate Solids Concentration		% w/w	25	25	25	25	5	Vendor	27	Undiluted cleaner flotation concentrate
	Concentrate Slurry Volume - Excl Spray Water		m³/h	237	326	326	263	6	Calculated		
	Concentrate SG			3.93	3.93	3.93	3.93	2	EPC	32	A
	Concentrate Wash Water Ratio (of Con m3/hr)			1.0	1.0	1.0	1.0	5	Vendor	27	A
	Cleaner Tail Solids Rate		t/h	183	145	159	203	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper
	Cleaner Tail Slurry Volumetric Flow		m³/h	945	910	957	1,038	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper
	Cleaner Tails Recycle		%	52.7	54.4	52.1	48.1	6	Calculated		The % of total tailings recycled back to feed hopper.
	Cleaner Tail Solids Rate		t/h	204	173	173	188	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper
	Cleaner Tail Slurry Volumetric Flow		m³/h	1,053	1,068	1,041	960	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	18	A
	Model			B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27	A
	Aspiration Type			Natural	Natural	Natural	Natural	5	Vendor	16	A
	Number Cells Installed			1	1	1	1	5	Vendor	18	A
	Number of Downcomers			24	24	24	24	5	Vendor	27	B
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27	A
	Froth Carry Rate	Nom	t/m²/h	3.50	4.81	3.50	3.51	5	Vendor	27	B
		Max	t/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27	A
		Calculated	t/m²/h	3.51	4.82	4.82	3.89	6	Calculated		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV		
				1	2	3	4						
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
7.5	Cleaner Scavenger Flotation			Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	1	Owner	66	Owner Supplied	B	
	Feed Streams			183	145	159	203	6	Calculated		New feed only		
	Solids Feed Rate	New Feed Only	t/h	380	300	317	385				Total feed to Jameson cell including tails recycle		
		Total	t/h	2,048	2,048	2,048	2,048	5	Vendor	27	Average of four modelled cases	C	
	Volumetric Feed Rate to Cell - Target	Total	m³/h	2,048	2,048	2,048	2,048	6	Calculated		Total feed to Jameson cell including tails recycle		
	Volumetric Feed Rate - Actual	Total	m³/h	15.6	12.6	13.2	15.8	6	Calculated		Total feed to Jameson cell including tails recycle		
	Feed Solids Concentration	Total	% w/w	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80		
	Particle Size p80	Design	µm										
	Mass Pull		%	7.6	13.3	12.1	7.6	6	Calculated				
	Concentrate Grade		% S	13.10	13.10	13.10	13.10	5	Vendor	19	Design Grade sulphur grade based on EPC circuit modelling	A	
	Concentrate Stage Recovery		% S	49.6	54.1	54.2	50.9	6	Calculated			A	
	Concentrate Recovery		% of Au in fl. feed	5.7	5.7	5.7	5.7	3	Assumed		Assumed to match the S recovery as seen in other areas of the flotation circuit	A	
	Concentrate Recovery		% of S in fl. feed	5.7	5.7	5.7	5.7	2	EPC	32		A	
	Concentrate Production		t/h	14	19	19	15	6	Calculated				
	Concentrate Solids Concentration		% w/w	25	25	25	25	2	EPC	32		B	
	Concentrate Slurry Volume - Excl Spray Water		m³/h	43	60	60	48	6	Calculated				
	Concentrate SG			3.25	3.25	3.25	3.25	2	EPC	32		A	
	Concentrate Wash Water Ratio (of Con m3/hr)			1	1	1	1	5	Vendor	27		A	
	Cleaner Scavenger Tail Solids Rate		t/h	169	125	140	188	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper		
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	948	914	960	1,041	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper		
	Cleaner Scavenger Tails Recycle		%	53.7	55.4	53.1	49.2	6	Calculated		The % of total tailings recycled back to feed hopper.		
	Cleaner Scavenger Tail Solids Rate		t/h	196	158	182	192	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper		
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	1,100	1,134	1,088	1,007	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper		
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	19		A	
	Model			B6500/24	B6500/24	B6500/24	B6500/24	1	Owner	27		A	
	Aspiration Type			Natural	Natural	Natural	Natural	1	Owner	16		A	
	Number Cells Installed			1	1	1	1	5	Vendor	19		A	
	Number of Downcomers			24	24	24	24	5	Vendor	27		B	
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27		A	
	Froth Carry Rate		t/m²/h	0.63	0.85	0.63	0.63	5	Vendor	27		B	
		Max	t/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27		A	
		Calculated	t/m²/h	0.63	0.86	0.86	0.70	6	Calculated				
8.0	CONCENTRATE HANDLING												
8.1	Concentrate Thickener (Brownfields)			Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)				Existing 30-TH-61		
	Feed Stream			Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	1	Owner	16		A	
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated				
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	230	319	319	256	6	Calculated				
	Feed Solids Concentration		% w/w	24.9	24.9	24.9	24.9	6	Calculated				
	Underflow Solids Concentration		% w/w	60	60	60	60	1	Owner	16		A	
	Testwork Flux Rate		t/m²/h	0.25	0.25	0.25	0.25	1	Owner	23		A	
	Calculated Flux Rate		t/m²/h	0.13	0.18	0.18	0.15	6	Calculated				
	Thickener Area Calculated		m²	573	573	573	573	6	Calculated				
	Diameter	Existing	m	27	27	27	27	1	Owner	16		A	
	Structural Design SG			3.8	3.8	3.8	3.8	1	Owner	16		A	
	Concentrate Thickener Sprays - Mill Circuit Water		m³/h	68.40	68.40	68.40	68.40	5	Vendor	113		1	
	8.2	Filtered Feed Tanks A & B (Brownfields)			Existing	Existing	Existing	Existing					
	Feed Streams			Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	1	Owner	43	Shared volume between Filtered Feed Tank A & B via balance line.	A	
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated				
Volumetric Feed Rate		m³/h	66	92	92	74	6	Calculated					
Feed Solids Concentration		% w/w	59.6	59.7	59.7	59.6	6	Calculated					
Number of Tanks			2	2	2	2	1	Owner	16		A		
Residence Time		h	21.2	15.3	15.3	19.0	6	Calculated					
Tank Capacity	per tank	m³	700	700	700	700	1	Owner	16		A		
8.3	Concentrate Filters (Brownfields)			Existing	Existing	Existing	Existing				Remains in operation to treat additional concentrate in excess of the UFG capacity.		
Feed Stream			Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	1	Owner	43		A		
Filter Type			Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	1	Owner	43		A		
Number of Filters			2	2	2	2	1	Owner	16		A		
Solids Feed Rate		t/h	0	0	0	0	6	Calculated			Solids have been prioritised through UFG circuits first.		
Volumetric Feed Rate		m³/h	0	0	0	0	6	Calculated					
Feed Solids Concentration	Target	% w/w	60.0	60.0	60.0	60.0	1	Owner	16		A		
	Actual	% w/w	0.0	0.0	0.0	0.0	6	Calculated					
Target Moisture Content		<% w/w	10.0	10.0	10.0	10.0	1	Owner	16		A		
Filter Area	per filter	m²	55.0	55.0	55.0	55.0	1	Owner	16		A		
Specific Filtration Rate	Nominal	m³/m²/h	0.14	0.14	0.14	0.14	1	Owner	16		A		
	Design	m³/m²/h	0.31	0.31	0.31	0.31	1	Owner	16		A		
8.4	Concentrate Stockpile			Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake				Remains in operation to treat additional concentrate in excess of the UFG capacity.		
Feed Stream			1	1	1	1	1	Owner	43		A		
No. of Stockpiles			1	1	1	1	1	Owner	16		A		
Concentrate Bulk Density		t/m³	2.2	2.2	2.2	2.2	1	Owner	16		A		
Angle of Repose	Nominal	Degrees	40.0	40.0	40.0	40.0	1	Owner	16		A		
Storage		days	2	2	2	2	1	Owner	16		A		
Capacity		m³	1,139	1,139	1,139	1,139	1	Owner	16		A		
TML for Bulk Shipment		% w/w	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	1	Owner	23		A		
9.0	CONCENTRATE ULTRAFINE GRINDING												
9.1	Concentrate UFG Mill 1 (Brownfields)			Existing	Existing	Existing	Existing				M3000		
	Circuit Feed Streams			Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	1	Owner	16, 80		1	
	Feed Rate - Fresh Feed	P80 9 µm	t/h	0.0	10.0	10.0	0.0	1	Owner	69		B	
		Design	t/h	0.0	13.0	13.0	0.0	1	Owner	16		A	
	Feed Solids Concentration		% w/w	45	45	45	45	1	Owner	4		A	
	Feed F80	Nominal	µm	144	144	144	144	1	Owner	16		A	
	Specific Energy	P80 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Owner	16		A	
	Type			IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	16		A	
	Model			M3000	M3000	M3000	M3000	1	Owner	16		A	
	Number			1	1	1	1	1	Owner	16		A	
	Configuration			Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	16		A	
	Mill Power Draw	Nominal	kW	860	860	860	860	1	Owner	16		A	
	Installed Motor Power		kW	1,120	1,120	1,120	1,120	1	Owner	16		A	
	UFG Mill Recycle Proportion		%	0	0	0	0	1	Owner	4		A	
	UFG Circuit Product Size P80	Nominal	µm	9.0	9.0	9.0	9.0	1	Owner	16		A	
	UFG Circuit Product Size P80 - Range		µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		A	
	Gland Seal Water		m³/h	0.0	1.2	1.2	0.0	1	Owner	2		A	
	Flushing Gland Seal Water		m³/h	0.0	5.4	5.4	0.0	1	Owner	2		A	
	9.2	UFG Mill 1 Trash Screen (Brownfields)			Existing	Existing	Existing	Existing					
	Feed Stream			UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	1	Owner	43		A	
	Type			1	1	1	1	1	Owner	16		A	
	Number of Screens per UFG Circuit		No.	1	1	1	1	1	Owner	16		A	
	Screen Solids Feed Rate per Screen	P80 9 µm	t/h	0	61	61	0	6	Calculated				
	Screen Feed Density	P80 9 µm	% w/w	45									

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
9.4	New Concentrate UFG Mill 2 & 3			Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)			M15,000, 2 off	A
	Circuit Feed Streams	UFG Mill 2/3						2	EPC		0
	Nominal Feed Rate - Fresh Feed	P ₈₀ 9 µm	t/h	30.0	30.0	30.0	30.0	1	Owner	24	A
	Maximum Feed Rate - Fresh Feed	P ₈₀ 12 µm	t/h	46.0	46.0	46.0	46.0	1	Owner	24	A
	Total Feed Solids Concentration		% w/w	47.5	47.5	47.5	47.7	5	Vendor	65	B
	Feed F ₈₀	Nominal	µm	144	144	144	144	1	Owner	16	A
	Specific Energy	P ₈₀ 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Owner	16	A
	Type			IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	23	A
	Model			M15,000	M15,000	M15,000	M15,000	1	Owner	23	A
	Number			2	2	2	2	1	Owner	23, 43	A
	Configuration			Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	23	A
	Mill Power Draw	Nominal	kW	3,500	3,500	3,500	3,500	1	Owner	5	A
	Installed Motor Power		kW	3,800	3,800	3,800	3,800	1	Owner	23	A
	UFG Mill Recycle Proportion		%	20	20	20	5	5	Vendor	65	B
	UFG Circuit Product Size P ₈₀	Nominal	µm	9.0	9.0	9.0	9.0	1	Owner	16	A
	UFG Circuit Product Size P ₈₀ - Range		µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16	A
	Gland Seal Water		m³/h	2.0	2.0	2.0	2.0	5	Vendor	65	B
	Flushing Gland Seal Water		m³/h	5.4	5.4	5.4	5.4	5	Vendor	65	B
9.5	UFG Mill 2 & 3 Trash Screens			Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen				A
	Feed Stream							1	Owner	23	B
	Type							1	Owner	74	A
	Number of Screens per UFG Circuit	No.		1	1	1	1	1	Owner	23	A
	Screen Solids Feed Rate per Screen	P ₈₀ 9 µm	t/h	37	46	46	42	6	Calculated		
	Screen Feed Density	P ₈₀ 9 µm	% w/w	59.6	59.0	59.0	59.6	6	Calculated		
	Screen Feed Flowrate per Screen	P ₈₀ 9 µm	m³/h	33	41	41	37	6	Calculated		
	Screen Size	L x W	m	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	5	Vendor	77	B
	Screen Area per Screen		m²	2.84	2.84	2.84	2.84	5	Vendor	77	B
	Specific Screen Flux Rate		m³/m²/h	14.8	17.7	17.7	16.1	6	Calculated		
	Screen Aperture		mm	1.0	1.0	1.0	1.0	1	Owner	16	A
	Screen Material			Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	5	Vendor	77	B
	Screen Spray Water		m³/h	9.0	9.0	9.0	9.0	5	Vendor	77	B
9.6	UFG Mill 2 & 3 Classification			Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX				A
	Cyclone Model			1	1	1	1	1	Owner	16	A
	Number of Cyclone Clusters per UFG Circuit			1	1	1	1	5	Vendor	54	A
	Number Installed Cyclones per Cluster	No.		24	24	24	24	5	Vendor	54	B
	Number of Cyclone Distributor Outlets Available	No.		24	24	24	24	5	Vendor	54	A
	Spare Outlets Available	No.		0	0	0	0	6	Calculated		
	Number of Operating Cyclones	No.		14	14	14	19	5	Vendor	54	B
	Number of Standby Cyclones	No.		10	10	10	5	6	Calculated		
	Cyclone Solids Feed Rate - per UFG circuit	P ₈₀ 9 µm	t/h	147.3	161.0	161.0	146.1	6	Calculated		
	Cyclone Feed Flow - per UFG circuit	P ₈₀ 9 µm	m³/h	319.0	348.5	348.5	315.1	6	Calculated		
	Cyclone Feed Solids Concentration		% w/w	32.8	32.8	32.8	32.9	5	Vendor	65	B
	Underflow Solids Concentration	P ₈₀ 9 µm	t/h	48.0	48.0	48.0	48.0	5	Vendor	65	B
	Underflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	109.9	115.0	115.0	104.4	6	Calculated		
	Overflow Solids Concentration	P ₈₀ 9 µm	% w/w	17.0	18.3	18.3	18.4	6	Calculated		
	Overflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	37.5	46.0	46.0	41.8	6	Calculated		
	Cyclone Circulating Load		%	293	250	250	250	5	Vendor	54	0
	Operating Pressure		kPa	247	247	247	248	5	Vendor	54	B
	Product Size P ₈₀		µm	9	9	9	9	1	Owner	5	A
9.7	Concentrate Pre-Leach Thickener (Brownfields)			Existing	Existing	Existing	Existing			Existing Pre-Leach Thickener 30-TH-31 Repurposed	
	Feed Streams			UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	1	Owner	24	A
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated		
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	362	475	475	368	6	Calculated		
	Feed Solids Concentration		% w/w	17	18	18	18	6	Calculated		
	Underflow Solids Concentration		% w/w	55	55	55	55	1	Owner	115	1
	Calculated Flux Rate		t/m²/h	0.13	0.18	0.18	0.15	6	Calculated		
	Testwork Flux Rate		t/m²/h	0.25	0.25	0.25	0.25	1	Owner	103	A
	Thickener Area Calculated		m²	573	573	573	573	6	Calculated		
	Diameter	Existing	m	27	27	27	27	1	Owner	103	A
	Thickener Spray Water		m³/h	15	15	15	15	3	Assumed		C
10.0	CIL - FLOTATION CONCENTRATE										
10.1	General Circuit (Brownfields)			Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3				
	Solids Feed Rate	Fresh	t/h	75	104	104.1	84	6	Calculated		
	Feed Pulp Density	Range	% w/w	50 - 60	50 - 60	50 - 60	50 - 60	1	Owner	108	0
	Actual Feed Pulp Density	Total	% w/w	49	49	48.7	49	6	Calculated		
	Slurry Feed Flow Rate	Fresh	m³/h	77	107	107	86	6	Calculated		
	Slurry Feed Flow Rate	Total	m³/h	99	137	137	110	6	Calculated		
	Leach Gold Extraction		%	90	90	90	90	9	Agreed		
	Adsorption Efficiency		%	99	99	99	99	1	Owner	44	B
10.2	Leaching (Brownfields)			Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)			CIL 2 repurposed for leach duty	
	Feed Streams							1	Owner	106	0
	Leach pH										
	30-TK-32A - 30-TK-32C			11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	1	Owner	108	0
	30-TK-32D			11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	1	Owner	108	0
	30-TK-32E			11 - 12.5	11 - 12.5	11 - 12.5	11 - 12.5	1	Owner	108	0
	Number of Tanks			7	7	7	7	1	Owner	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Owner	16	A
	Total Tank Capacity		m³	11,795	11,795	11,795	11,795	6	Calculated		
	Residence Time		h	119	86	107	107	6	Calculated		
	Minimum Residence Time		h	80	80	80	80	1	Owner	103	A
	Oxygen Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	16	A
	Oxygen Addition Method			Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	1	Owner	43	4
	Dissolved Oxygen Concentration	Target	ppm	10	10	10	10	1	Owner	16	A
	Cyanide Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	43	A
	Cyanide Concentrations										
	30-TK-32A	Target	ppm	500	500	500	500	1	Owner	108	0
	30-TK-32B	Target	ppm	1,000	1,000	1,000	1,000	1	Owner	108	0
	30-TK-32C	Target	ppm	2,200	2,200	2,200	2,200	1	Owner	108	0
	30-TK-32D	Target	ppm	5,000	5,000	5,000	5,000	1	Owner	108	0
	30-TK-32E	Target	ppm	6,500	6,500	6,500	6,500	1	Owner	108	0
	Milk of Lime Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43	A
	Lead Nitrate Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43	A
10.3	Adsorption (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3			CIL 3 repurposed for adsorption duty	
	Number of Tanks			6	6	6	6	1	Owner	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Owner	16	A
	Total Adsorption Tank Live Volume		m³	10,110	10,110	10,110	10,110	6	Calculated		
	Residence Time		h	102	74	92	92	6	Calculated		
	Minimum Residence Time		h	60	60	60	60	1	Owner	103	A
	Total Leach & Adsorption Time		h	221	160	160	199	6	Calculated		
	Carbon Concentration	per tank	g/L	20	20	20	20	1	Owner	16	A
	Maximum Carbon Residence Time		days	14	14	14	14	1	Owner	103	A
	Carbon Movements per Day	Max	t/d	33	33	33	33	6	Calculated		
	Carbon In Circuit	Total	t	202.2	202.2	202.2	202.2	6	Calculated		
	Design Carbon Loading		g/t	4,490	4,490	4,490	4,490	1	Owner	16, 103	A
	Carbon Transfer Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16	A
	Carbon Transfer Period	per cycle	h	4.0	4.0	4.0	4.0	6	Calculated		
	Carbon and Pulp Transfer Rate		m³/h	138	138	138	138	2	EPC	119	4

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
				1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P90 220um					
10.4	Loaded Carbon Recovery Screen & Transfer			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			
	Loaded Carbon Advance Method			275	275	275	275	2	EPC			
	Loaded Carbon Pump & Pulp Forwarding Rate			2	2	2	2	2	EPC			
	Loaded Carbon Transfer Time	Pump to Screen	m³/h h									
	Feed Streams			Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	1	Owner	As per P3-EPC-COR-602		
	Type			Linear Motion	Linear Motion	Linear Motion	Linear Motion	1	Owner	New Derrick Linear Motion Screen as per P3-EPC-COR-602		
	Number	No.		1	1	1	1	1	Owner	New Derrick Linear Motion Screen as per P3-EPC-COR-602		
	Feed Rate		m³/h	275	275	275	275	6	Calculated			
	Area		m²	3.80	3.80	3.80	3.80	5	Vendor	117		
	Deck Type			Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	5	Vendor	117		
	Aperture		mm	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	5	Vendor	117		
	Total Spray Water Addition		m³/h	10	10	10	10	3	Assumed			
	Loaded Carbon Moisture		% w/w	40	40	40	40	1	Owner			
	Loaded Carbon Transfer Moisture	Maximum	% w/w	55	55	55	55	1	Owner	4	Screen Oversize	
	Loaded Carbon Transfer Hopper Capacity	Minimum	m³	23	23	23	23	2	EPC			
	Loaded Carbon Transfer Time	Hopper to Column	min	60	60	60	60	2	EPC			
	10.5	Interstage Carbon Screens (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3				
		Type			Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	16	
Supplier & Model				Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	1	Owner	16		
Number		per tank	No.	1	1	1	1	1	Owner	16		
Screen Feed Rate - per Screen		w/o carbon advance	m³/h	102	140	140	113	6	Calculated			
		w/ carbon advance	m³/h	240	278	278	251	6	Calculated			
Area Selected		Selected	m²	12.0	12.0	12.0	12.0	1	Owner	16		
Specific Screen Flux Rate		w/ carbon advance	m³/m²/h	20.0	23.1	23.2	20.9	6	Calculated			
		Maximum	m³/m²/h	100	100	100	100	1	Owner	16		
Screen Material				Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	102		
Screen Aperture			mm	1.0	1.0	1.0	1.0	1	Owner	16		
10.6		CIL 3 Tails Hopper										
	Feed Flowrate		m³/h	106	144	144	118	6	Calculated			
	Discharge Destination			CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43		
	Cyanide Concentration		ppm	750	750	750	750	1	Owner	108		
11.0 FLOTATION TAILINGS THICKENING												
11.1	Flotation Tailings Thickener			Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails			Owner Supplied		
	Feed Stream							1	Owner	16		
	Solids Feed Rate		th	3,071	2,858	3,042	3,411	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	6,871	6,492	6,884	6,187	6	Calculated			
	Feed Solids Concentration - Excl. Diluted Flocculant		% w/w	33.0	32.6	32.7	38.9	6	Calculated			
	Underflow Solids Concentration	Nominal	% w/w	55.9	56.0	56.0	55.9	6	Calculated		Achieve 55 % solids concentration in flotation tails CIL circuit post dilution (i.e. total feed % solids)	
	Underflow Solids Concentration Range		% w/w	50-62	50-62	50-62	50-62	1	Owner	23		
	Testwork Flux Rate		t/m²/h	1.50	1.50	1.50	1.50	1	Owner	23		
	Operating Flux Rate Range		t/m²/h	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1	Owner	16		
	Nominal Flux Rate		t/m²/h	1.09	1.01	1.08	1.21	6	Calculated			
	Thickener Area Calculated		m²	2,827	2,827	2,827	2,827	6	Calculated			
	Diameter Selected		m	60	60	60	60	1	Owner	16		
	Thickener Underflow Destination(s)			CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43		
	Number of Underflow Discharge Pumps			2	2	2	2	1	Owner	16		
	Pump Configuration			Duty/Standby	Duty/Standby	Duty/Standby	Duty/Standby	1	Owner	43		
	Total Thickener Underflow		m³/h	3,329	3,087	3,289	3,701	6	Calculated			
	12.0 CIL - FLOTATION TAILINGS											
	12.1	General Circuit			CIL 4	CIL 4	CIL 4	CIL 4				
Feed Streams				Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	1	Owner	106		
Number of Trains				1	1	1	1	1	Owner	43		
Fresh Solids Feed Flow Rate			th	3,151	2,969	3,153	3,500	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
Total Solids Feed Flow Rate			th	3,164	2,982	3,166	3,515	6	Calculated			
Fresh Slurry Feed Flow Rate			m³/h	3,438	3,234	3,437	3,822	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
Total Slurry Feed Flow Rate			m³/h	3,511	3,305	3,510	3,900	6	Calculated			
Target Feed Pulp Density		Target	% w/w	55	55	55	55	1	Owner	16, 103	Post dilution (i.e. total feed % solids)	
Maximum Feed Pulp Density			% w/w	60	60	60	60	1	Owner	43	Post dilution (i.e. total feed % solids)	
Feed Pulp Density		Actual	% w/w	55	55	55	55	6	Calculated		Post dilution (i.e. total feed % solids)	
Leach Gold Extraction			%	38.0	38.0	38.0	38.0	1	Owner	44		
Adsorption Efficiency			%	88.0	88.0	88.0	88.0	1	Owner	44	Range 84 - 92%	
12.2		Leaching			CIL 4	CIL 4	CIL 4	CIL 4				
		Leach pH			8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	1	Owner	108	
		F21-TK-411			2	2	2	2	1	Owner	43	
		Number of Tanks	per train		2	2	2	2	1	Owner	43	
		Live Tank Capacity		m³	5,600	5,600	5,600	5,600	1	Owner	43	
		Total Tank Capacity		m³	11,200	11,200	11,200	11,200	6	Calculated		
	Residence Time		h	3.2	3.4	3.2	2.9	6	Calculated			
	Oxygen Addition Points			All Leach Tanks	All Leach Tanks	All Leach Tanks	All Leach Tanks	1	Owner	43		
	Oxygen Addition Method			Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	1	Owner	43		
	Dissolved Oxygen Concentration	Target	ppm	15	15	15	15	1	Owner	108		
	Cyanide Addition Points			Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	1	Owner	43		
	Cyanide Concentration		ppm	175	175	175	175	1	Owner	108		
Milk of Lime Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43			
12.3	Adsorption			CIL 4	CIL 4	CIL 4	CIL 4					
	Number of Tanks	per train		6	6	6	6	1	Owner	43	Provision in the layout for a future, 7th adsorption tank	
	Live Tank Capacity		m³	5,600	5,600	5,600	5,600	1	Owner	43		
	Total Adsorption Tank Live Volume		m³	33,600	33,600	33,600	33,600	6	Calculated			
	Residence Time		h	9.6	10	10	9	6	Calculated			
	Total Leach & Adsorption Time		h	12.8	13.6	12.8	11.5	6	Calculated		2 Leaching & 6 Adsorption Tanks will not achieve the 14 hours total residence time. Client is aware and has advised to maintain 8 tanks with the 9th as a future.	
	Oxygen Addition Points			First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	1	Owner	43		
	Oxygen Addition Method			Side Lances	Side Lances	Side Lances	Side Lances	1	Owner	43		
	Carbon Concentration - Duty	per tank	g/L	12.5	12.5	12.5	12.5	1	Owner	109		
	Carbon Concentration Used for Carbon Transfer		g/L	12.5	12.5	12.5	12.5	1	Owner	109		
	Carbon Movements per Day	Max	t/d	30	30	30	30	6	Calculated			
	Carbon Residence Time	Maximum	d	14	14	14	14	1	Owner	103		
	Carbon In Circuit	@ 8 g/L	t	269	269	269	269	6	Calculated			
		@ 10 g/L	t	336	336	336	336	6	Calculated			
		@ 12.5 g/L	t	420	420	420	420	6	Calculated			
	Nominal Carbon Loading		g/t	461	461	461	461	1	Owner	16, 103		
	Carbon Loading Range		g/t	332 - 579	332 - 579	332 - 579	332 - 579	1	Owner	16, 103		
	Carbon Transfer Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			
Carbon Transfer Period		h	5.5	5.5	5.5	5.5	6	Calculated				
Carbon and Pulp Transfer Rate		m³/h	220	220	220	220	2	EPC				
Loaded Carbon Advance Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC				
Loaded Carbon Pump & Pulp Forwarding Rate		m³/h	280	280	280	280	2	EPC				
Loaded Carbon Transfer Time		h	4.3	4.3	4.3	4.3	6	Calculated				
12.4	Loaded Carbon Recovery Screen			CIL 4	CIL 4	CIL 4	CIL 4					
	Feed Streams			First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	1	Owner	16		
	Type			Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	1	Owner	16		
	Number	No.		1	1	1	1	1	Owner	16		
	Feed Rate		m³/h	280	280	280	280	6	Calculated			
	Selected Size	Width	m	1.83	1.83	1.83	1.83	5				

[illegible]

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
14.3	Acid Washing										
	Acid Washes Per Week	No. t		12			6	Calculated			
	Acid Wash Column Capacity			15			1	Owner	8	Combined Acid Wash & Elution column	A
	Acid Type			Hydrochloric Acid			1	Owner	103		A
	Acid Dosing Method			Pumped			1	Owner	103		A
	Acid Concentration	% w/v HCl		3			1	Owner	103		A
	Acid Injection Time	min		17			1	Owner	103		A
	Acid Soak Time	min		60			1	Owner	103		A
	Acid Water Rinse Time	min		120			1	Owner	103		A
	Total Acid Wash Solution Flowrate	BV/h		2			1	Owner	8		A
	Total Acid Wash Solution Flowrate	m³/h		63.8			6	Calculated			
	Dilute Acid Flowrate	BV		0.6			6	Calculated			
	Dilute Acid Solution Used	m³		63.8			6	Calculated			
	Neat Acid Solution Used	m³		18.085			6	Calculated			
	Waste Acid Solution Destination			1.5			6	Calculated			
	Acid Wash Temperature	°C		Final Tails Tank			1	Owner	43		B
				Ambient			1	Owner	44		A
14.4	Water Rinse										
	Water Flowrate	m³/h		63.8			6	Calculated		Based on 2 BV/h	
	Acid Water Rinse Time	min		120			1	Owner	103		A
	Volume Water Used	m³		128			6	Calculated			
	No. Bed Volumes	BV		4			1	Owner	8, 103		
	Solution Destination			Final Tails Tank			1	Owner	43		B
	Acid Rinse Temperature	°C		Ambient			1	Owner	44		A
	Rinse Water Source			Scheme Water			1	Owner	103		A
14.5	Elution										
	Elution Type			Split AARL			1	Owner	43		A
	Required Elutions Per Week	No. t		12			1	Owner	8, 43		B
	Elution Column Capacity			15			1	Owner	8, 43		A
	Carbon Bed Volume	m³		31.9			6	Calculated			
14.6	Elution Column Operating Pressure	Top of Column kPa(g)		340			2	EPC		To prevent flashing (boiling) of solution inside column.	B
	Pre-treatment (with Lean Eluate)										
	Cycle Time	min		30			2	EPC			A
	Solution Flowrate	BV/h		2			1	Owner	8		A
	Temperature	°C		63.8			6	Calculated			
	No. of Bed Volumes			95			2	EPC			A
	Pre-soak Solution Volume	m³		1.0			6	Calculated		5 BVs of Pregnant Eluate required by owner (Source 8)	
	Column Discharge Destination			31.9			6	Calculated			
	Solution Strength	% w/v NaCN		Lean Eluate Tank			8	Industry			B
				1.7			1	Owner	103		A
	Sodium Cyanide Consumption	kg/pre-treat		2			1	Owner	103		A
	Cautslic Consumption	Neat Solution m³/pre-treat		543			6	Calculated			
	Lean Eluate Solution Make-up Per Mix	Neat Solution m³/pre-treat		1.51			6	Calculated			
				638			6	Calculated			
14.7	Elution (with Lean Eluate)										
	Elution Cycle Time	min		0.85			6	Calculated			
	Elution Solution Flowrate	BV/h		29.5			6	Calculated			
	Eluate Consumed	m³		150			2	EPC			A
	Elution Temperature	°C		2			1	Owner	8		A
	Column Discharge Destination			63.8			6	Calculated		5 BVs of Pregnant Eluate required by owner (Source 15)	
				5.0			6	Calculated			
				159.6			6	Calculated			
				130			2	EPC			A
				CIL4 Eluate Circulation Tanks			8	Industry			A
14.8	Elution & Cooling (with Scheme Water)										
	Cycle Time - Scheme Water (with heaters)	min		90			6	Calculated			
	Elution Temperature	°C		130			2	EPC			C
	Water Flowrate	BV/h		2			1	Owner	8		A
	Water Consumed	m³		63.8			6	Calculated			
				3.0			6	Calculated			
				95.7			6	Calculated			
	Cycle Time - Scheme Water (no heaters, cooling)	min		60			2	EPC			A
	Elution Cookdown Temperature	°C		-80			2	EPC			A
	Water Flowrate	BV/h		2			1	Owner	8		A
	Water Consumed	m³		63.8			6	Calculated			
				2.0			6	Calculated			
				63.8			6	Calculated			
	Total Cycle Time	min		150			2	EPC			A
	Water Flowrate	BV/h		2			1	Owner	8		A
	Total Water Consumed	m³		63.8			6	Calculated		Total of 5 BVs of Lean Eluate required by owner (Source 8)	
				5.0			6	Calculated			
				159.6			6	Calculated			
	Column Discharge Destination			CIL4 Eluate Circulation Tanks and Lean Eluate Tank			8	Industry		When the CIL4 eluate circulation tank is full, discharge switches to the lean eluate tank	A
	Carbon Transfer In	min		30.0			6	Calculated			
	Total Acid Time (including rinsing)	h		0.5			6	Calculated			
		min		197			6	Calculated			
		h		3.3			6	Calculated			
	Total Elution Time (including Cooling)	min		330			6	Calculated			
		h		5.5			6	Calculated			
	Carbon Transfer Out	min		53.5			6	Calculated			
		h		0.9			6	Calculated			
	Total	h		10.2			6	Calculated			
14.9	Heating										
	Elution Heater Type			LPG Fired Elution Heater			1	Owner	43		A
14.10	Lean Eluate Tank										
	Total Volume of Elution Eluate Per Strip	Pregnant m³		160			6	Calculated			
	Number of Lean Eluate Tanks			1			1	Owner	43		A
14.11	Elution Water Tank										
	Total Live Volume Elution Water Tank	Minimum m³		160			6	Calculated			
14.12	Eluate Expansion Tank										
	Number of Eluate Expansion Tanks			1			2	EPC			B
14.13	Total Water Consumption Per Strip										
	Carbon Transfer	Scheme m³/strip		8			6	Calculated			
14.14	Acid Wash	Scheme m³/strip		17			6	Calculated			
	Water Rinse	Scheme m³/strip		128			6	Calculated			
	Water Elution & Cooling	Scheme m³/strip		160			6	Calculated			
	Carbon Eduction (Elution to Regen)	Scheme m³/strip		60			6	Calculated			
	Carbon Quench to Sizing Screen	Scheme m³/strip		60			6	Calculated			
	Total Per Strip	Scheme m³/strip		432			6	Calculated			
	Elution Electrowinning										
	Number Circuits			1			2	EPC			A
15.0	Goldroom Operation (Brownfields)										
	Feed Sources			Existing			1	Owner	43		A
	Sludge Filter Operating Days	days/week		Flotation Con EW product (CIL2/3) & Flotation Tails EW product (CIL 4)			1	Owner	16		A
	Sludge Dewatering Cone Capacity	m³		2			1	Owner	16		A
	Sludge Filter Make			1.3			1	Owner	16		A
	Sludge Filter Model			TEFSA			1	Owner	16	470x470mm plate filter, 6 bar design pressure	A
				HPLE			1	Owner	16		
	Mercury Retort Oven Operating Days	days/week		2			1	Owner	16		A
	Mercury Retort Oven Make			Locthead			1	Owner	16		A
	Mercury Retort Oven Model			F66OV66			1	Owner	16	20 ft³ electric powered mercury retort	A
	Calcine per week	kg		800 - 1,200			1	Owner	16	Existing, Calcine logs, Max Capacity 1,600 kg per cycle	A
	Smelting Furnace Operating Days	days/week		2			1	Owner	16		A
	Furnace Make			Furnace Engineering			1	Owner	16		A
	Furnace Model			Westofen KBU 80/H			1	Owner	16	2000 L x 3000 D x 1710 H, Weishaupt G3-1a burner	A
	Furnace Extraction Fan			75 kW / 1,480 RPM			1	Owner	16	IP55, 415V with 315xSSPB motor pulley and 212xSSPB fan pulley	A
	Furnace Scrubber			Metalcraft SN50247/1			1	Owner	16		A
	Smelts per week			6			1	Owner	16	Two downloads per week, three pours per day	A

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
16.0 CARBON REGENERATION - FLOTATION CONCENTRATE											
16.1	Barren Carbon Transfer (Brownfields)			Existing							
	Transfer Water Source			Carbon Transfer Water (Scheme Water)			1	Owner	16, 103		A
	Carbon Transfer Time	min		30			1	Owner	16, 103		A
	Carbon Transfer Concentration	% w/w		20			1	Owner	57		B
	Transfer Water Flowrate	m³/h		88			6	Calculated		Based off 11t elution column	
	Transfer Water Required	m³		44			6	Calculated		Based off 11t elution column	
16.2	Regeneration Kiln Dewatering Screens (Brownfields)			Existing							
	Type			Horizontal Vibrating Screen			1	Owner	16, 103		A
	Number	No.		3			1	Owner	16, 103	1x Per Regeneration Kiln	A
	Specific Screen Flux Rate	t/m²/h	Max Recommended	4			1	Owner	16, 103		A
	Selected Size	m	Width	1.2			1	Owner	16, 103		A
		m	Length	2.4			1	Owner	16, 103		A
	Area	m²		2.9			6	Calculated			
	Feed % Solids	% w/w		20			1	Owner	57		B
	% Mass Split to Oversize	%		100			2	EPC			B
	Oversize Moisture %	% w/w		60			1	Owner	57		B
16.3	Carbon Regeneration Kiln (Brownfields)			Existing							
	Regeneration Kiln Type			Horizontal			1	Owner	16, 103		A
	Regeneration Kiln Model			Metso RK600/650HR			1	Owner	16, 103		A
	Number			3			1	Owner	16, 103		A
	Eluted Carbon to be Regenerated	%		100			1	Owner	16, 103		A
	Regenerated Carbon per Batch	t		11			6	Calculated			
	Temperature	°C		750			1	Owner	16, 103		A
	Required Kiln Operating time	h/batch		20			1	Owner	100		A
	Capacity - Nominal	kg/h		550			6	Calculated			
	Capacity - Design	kg/h		650			1	Owner	100		A
16.4	Carbon Quench Tank (Brownfields)			Existing							
	Carbon Quench Tank Volume	m³	Each	12			1	Owner	16, 103	3 off Quench Tanks	A
	Carbon Transfer Time	min		30			1	Owner	16, 103		A
	Carbon Transfer Concentration	% w/w		20			1	Owner	57		B
	Transfer Water Flowrate	m³/h	Each	88			6	Calculated		Based off 11t elution column	
	Transfer Water			Carbon Transfer Water (Scheme Water)			1	Owner	16, 103		B
	Transfer Water Required	m³		44			6	Calculated			
16.5	CIL 3 Barren Carbon Sizing Screen			New							
	Type			Horizontal Vibrating Screen			1	Owner	100		A
	Number	No.		1			1	Owner	100		A
	Feed Rate	m³/h		98			6	Calculated		Based off 1 x 11t elution columns	
	Feed Rate	t/h		22			6	Calculated		Based off 1 x 11t elution columns	
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.60			6	Calculated			
	Specific Screen Flux Rate	m³/m²/h	Actual	17.5			6	Calculated		w/o screen spray water	
	Deck Type			Polyurethane			1	Owner	100		A
	Aperture	mm		1.25 x 18.5			5	Vendor	84		1
	% Mass Split to Oversize	%		100			2	EPC	84		B
	Oversize Moisture %	% w/w		40			1	Owner	4		B
	Screen Spray Water	m³/h		13			5	Vendor			0
16.6	Regeneration Kiln Off-Gas Treatment (Brownfields)			Existing							
	Venturi Scrubber			1			1	Owner	16, 103		A
	Number			375			1	Owner	16, 103		A
	Design Dust Load	mg/Nm³		546			1	Owner	16, 103		A
	Design Mercury	g/h		831			1	Owner	16, 103		A
	Design VOC	g/h		250			1	Owner	16, 103		A
	Design Gas Temperature	°C		Process			1	Owner	16, 103		A
	Water Type										
	Packed Bed Wet Scrubber			Existing							
	Number			1			1	Owner	16, 103		A
	Design Gas Flowrate	Nm³/h		6,481			1	Owner	16, 103		A
	Design Gas Temperature	°C		80			1	Owner	16, 103		A
	Water Type			Process			1	Owner	16, 103		A
	Mist Eliminator			Existing							
	Number			1			1	Owner	16, 103		A
	Design Gas Flowrate	Nm³/h		3,916			1	Owner	16, 103		A
	Design Gas Temperature	°C		58			1	Owner	16, 103		A
	Water Type			Scheme			1	Owner	16, 103		A
	Regenerative Thermal Oxidiser			Existing							
	Number			1			1	Owner	16, 103		A
	Design Gas Flowrate	Nm³/h		4,132			1	Owner	16, 103		A
	Design Gas Temperature	°C		58			1	Owner	16, 103		A
	Carbon Filters			Existing							
	Number			2 (Duty / Standby)			1	Owner	16, 103		A
	Design Gas Flowrate	Nm³/h		4,503			1	Owner	16, 103		A
	Design Gas Temperature	°C		80			1	Owner	16, 103		A
	Design Mercury Loading	g/h		387			1	Owner	16, 103		A
	Stack Emission Limits										
	Mercury	ppmv		0.20			1	Owner	16, 103	FIM ERP Commissioning Report	A
	VOC	ppmv		3.2			1	Owner	16, 103		
17.0 CARBON REGENERATION - FLOTATION TAILINGS											
17.1	Barren Carbon Transfer			New - Flotation Tailings							
	Transfer Water Source			Carbon Transfer Water (Scheme Water)			1	Owner	103		B
	Carbon Transfer Time	min		53			2	EPC			0
	Carbon Transfer Concentration	%w/w		20			1	Owner	57		B
	Transfer Water Flowrate	m³/h		67			6	Calculated		Based off 1x 15t carbon batch	
	Transfer Water Required	m³		60			6	Calculated		Based off 1x 15t carbon batch	
17.2	Regeneration Kiln Dewatering Screen			New							
	Type			Horizontal Vibrating Screen			5	Vendor	84		A
	Number	No.		1			1	Owner	103	One elution column	A
	Feed Flow Rate	m³/h		75			6	Calculated			
	Solids Feed Rate	t/h		17			6	Calculated			
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.6			6	Calculated			
	Specific Screen Flux Rate	t/m²/h	Maximum	4			1	Owner			
		m³/m²/h	Actual	13.4			6	Calculated	103		B
		t/m²/h	Actual	3.0			6	Calculated			
	Deck Type			Polyurethane			5	Vendor	84		B
	Aperture	mm		1.2 x 18.5			5	Vendor	84		0
	Feed % Solids	% w/w		20			1	Owner	57		B
	% Mass Split to Oversize	%		100			2	EPC			B
	Oversize Moisture %	% w/w	Nominal	40			2	EPC	52		B
		% w/w	Maximum	60			1	Owner	57		B
	Screen Spray Water	m³/h		13			5	Vendor			0
17.3	Carbon Regeneration Kiln			New							
	Reactivation Kiln Type			Horizontal - Indirect Fired, Electric			1	Owner	103		B
	Number			1			1	Owner	103		A
	Regenerated Carbon per Batch	t		30			6	Calculated		2 strips per day from 15t elution column	
	Regen Kiln Feed Hopper Capacity	%	% of full column size	200			1	Owner	103		B
	Regen Kiln Feed Hopper Capacity	t		30			6	Calculated			
	Required Kiln Operating time	h/batch	Nominal	20			8	Industry			A
	Capacity - Nominal	kg/h		1,650			1	Owner	43		A
	Operating Temperature	°C		650 - 750			1	Owner	102		A
	Design Operating Temperature	°C		750			1	Owner	103		
	Fraction of Carbon Regenerated	%		100			1	Owner	103		
	Regenerated Carbon Mercury Grade	Hg ppm	Nominal	1.3			1	Owner	16		B
		Hg ppm	Range	0 - 4			1	Owner	16		B
17.4	Carbon Quench Tank			New							
	Carbon Quench Tank Volume	m³	Minimum	20			1	Owner			A
	Carbon Transfer Time	min		53			2	EPC			0
	Carbon Transfer Concentration	% w/w		20			2	EPC			C
	Transfer Water Flowrate	m³/h		67.3			6	Calculated		Based off one 15t elution column	
	Transfer Water			Quench Hopper filled with Scheme Water			1	Owner	100		0
	Transfer Water Required	m³		Carbon Transfer completed with Raw Water			6	Calculated		Based off one 15t elution column	
17.5	CIL 4 Carbon Sizing Screen			New							
	Type			Horizontal Vibrating Screen			1	Owner	100		A
	Number	No.		1			1	Owner	103	Single CIL4 Train	A
	Feed Flow Rate	m³/h		75			6	Calculated			
	Solids Feed Rate	t/h		17			6	Calculated			
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.6			6	Calculated			
	Specific Screen Flux Rate	m³/m²/h	Maximum	35.0			1	Owner	103		B
		t/m²/h	Actual	3.0			6	Calculated		w/o screen spray water	
		m³/m²/h	Actual	13.4			6	Calculated			
	Deck Type			Polyurethane			5	Vendor	84		B
	Aperture	mm		1.25 x 18.5			5	Vendor	84		1
	% Mass Split to Oversize	%		100			2	EPC	84		B
	Oversize Moisture %	% w/w		40			1	Owner	4		B
	Screen Spray Water	m³/h		13			5	Vendor			0

AREA	PARAMETER		UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV		
				1	2	3	4							
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um							
17.6	Regeneration Kiln Off-Gas Treatment				New									
	Mercury Recovery	Target	%		>92			1	Owner	43		B		
	Design Mercury	New - Range	g/t		20 - 72			1	Owner	16	Brownfields barren carbon mercury grade CIL2/3 20th - 80th Percentile (2017 - 2023).	B		
		New - Design	g/h		108			6	Calculated		Based on the upper limit of brownfields barren carbon mercury grade. 72g/t for 1,500 kg carbon/h through regen kiln			
	VOC	New	g/h		639			6	Calculated		Based on brownfields design VOC of existing Venturi Scrubber. Assumes 3x existing Carbon Regen Kilns running at design throughput. Estimate only - stack emission criteria used as basis of design			
		New - Nominal	°C		316			5	Vendor	52		B		
		New - Design	°C		400			5	Vendor	52		1		
		New - Nominal	Nm³/h		1,335			6	Calculated	52	Based on 40% moisture CIL4 Carbon Regeneration Kiln feed	B		
	Gas Flow	New - Design	Nm³/h		3,003			6	Calculated	52	Based on 60% moisture CIL4 Carbon Regeneration Kiln feed	B		
	Carbon Fines Knock-Out Box													
	Number				1			5	Vendor	52		A		
	Dimensions		m		1.22 x 1.8			5	Vendor	52		1		
	Water Type	Sprays			Scheme			5	Vendor	52		A		
	Water Flowrate		m³/h		8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4		
	Kiln Off-Gas Condenser													
	Number				1			5	Vendor	52		A		
	Cooler Type				Shell and Tube			5	Vendor	52		1		
	Water Type				Cooling Water (Scheme)			5	Vendor	52		A		
	Kiln Mist Eliminator				New									
	Number				1			5	Vendor	52		A		
	Dimensions		m		1.22 x 1.8			5	Vendor	52		1		
	Water Type	Sprays			Scheme			5	Vendor	52		A		
	Water Flowrate		m³/h		8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4		
	Kiln Duct Heater													
	Number				1			5	Vendor	52		A		
	Power		kW		7.5			5	Vendor	52		1		
	Kiln Adsorption Vessel													
	Number				1			5	Vendor	52		A		
	Type				SIC Bed - CGS Dual Banjo			5	Vendor	52		B		
	Dimensions		m		1.98 x 2.7			5	Vendor	52		1		
	Carbon Type & Size		mm		3mm Mersorb HT			5	Vendor	52		1		
	Kiln Exhaust Fan													
	Number				1			5	Vendor	52		A		
	Design Duty		Am³/h		3,003			5	Vendor	52	At normal conditions of: 1 atm and 20 deg C	1		
	Power		kW		12.5			5	Vendor	52		1		
	Kiln Discharge Stack													
	Number				1			1	Owner	52		A		
	Height		m		35			1	Owner	52		B		
	Stack Emission Limits													
	Mercury	Maximum	ppmv		0.20			1	Owner	43	At standard temperature and pressure	A		
			micrograms/m³		1,800			1	Owner	52		B		
	Mercury	Vendor System	ppmv		0.027			6	Calculated			A		
VOC	Maximum	ppmv		3.20			1	Owner	43		A			
18.0 CYANIDE DESTRUCT														
	Feed Streams	1			Kaltails Decant	Kaltails Decant	Kaltails Decant	Kaltails Decant	1	Owner	35	TBC with updated drawings by others	2	
		2			offsite Decant & FIM I Decant	offsite Decant & FIM I Decant	offsite Decant & FIM I Decant	offsite Decant & FIM I Decant	1	Owner	35	TBC with updated drawings by others	2	
		3			Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	1	Owner	35	TBC with updated drawings by others	2	
	Decant Water Recovered from Tailings	% of Total Tailings Water	%		49	49	49	49	1	Owner	35		A	
			m³/h		728	728	728	728	1	Owner	44		B	
	Decant Water from Kaltails		m³/h		1,940	1,872	1,941	2,071	6	Calculated				
	Total Feed Flow to Cyanide Water Tank				Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	1	Owner	35		A	
	Method				<1.5	<1.5	<1.5	<1.5	1	Owner	35		A	
	Residual WAD Cyanide Target	Stage 2			Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	1	Owner	35		2	
	Dosing Points	Location			3	3	3	3	1	Owner	35		2	
	Dosing Method	Number			Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	1	Owner	35	Standby pump acts as a lag pump to the duty operating pumps in the event of high peroxide demand	2	
	Dosing Control Method				Online CN _{WAD} Analyser	Online CN _{WAD} Analyser	Online CN _{WAD} Analyser	Online CN _{WAD} Analyser	1	Owner	35		A	
19.0 REAGENTS														
19.1	Activator - Copper Sulphate													
	Delivery Form		L		Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64		A	
	Delivery Size per Trailer				24,000	24,000	24,000	24,000	5	Vendor	112		1	
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A	
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated				
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64		A	
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		B	
	Solution Mix Strength (as CuSO ₄ .5H ₂ O)		% w/v		25	25	25	25	1	Owner	16		A	
	Solution Mix Strength (as CuSO ₄)		% w/v		16	16	16	16	1	Owner	64		A	
	Supply Solution SG		t/m³		1.17	1.17	1.17	1.17	1	Owner	16		A	
	Storage Volume	Minimum	# deliveries		2.5	2.5	2.5	2.5	1	Owner	83		B	
	Number of Storage Tanks				1	1	1	1	1	Owner	16		A	
	Live Volume per Tank		m³		120	120	120	120	6	Calculated			B	
	Residence Time	Actual	d		11	12	11	10	6	Calculated		Residence time calculated using nominal consumption rates.		
	Consumption													
	SAG Mill Discharge	Nominal	g/t fresh feed to Cyclone Feed Hopper		35	35	35	35	1	Owner	23, 75	Added at Cyclone Feed Hopper	A	
	Cleaner Scavenger	Nominal	g/t fresh cleaner scav feed		10	10	10	10	1	Owner	23, 75		A	
	SAG Mill Discharge	Design	g/t fresh feed to Cyclone Feed Hopper		70	70	70	70	6	Calculated		Added at Cyclone Feed Hopper. 100% Design Factor for pumping.		
	Cleaner Scavenger	Design	g/t fresh cleaner scav feed		20	20	20	20	6	Calculated		100% Design Factor for pumping.		
	SAG Mill Discharge	Nominal	L/h		440	414	440	489	6	Calculated				
	Cleaner Scavenger	Nominal	L/h		7.3	5.8	6.3	8.1	6	Calculated				
	SAG Mill Discharge	Design	L/h		881	829	881	978	6	Calculated		100% Design Factor for pumping.		
	Cleaner Scavenger	Design	L/h		14.7	11.6	12.7	16.3	6	Calculated		100% Design Factor for pumping.		
	Nominal Consumption	as CuSO ₄ .5H ₂ O	L/h		447	414	440	489	6	Calculated				
		as CuSO ₄ .5H ₂ O	m³/h		0.45	0.41	0.44	0.49	6	Calculated				
	Design Consumption	as CuSO ₄ .5H ₂ O	L/h		896	841	894	995	6	Calculated				
		as CuSO ₄ .5H ₂ O	m³/h		0.90	0.84	0.89	0.99	6	Calculated				
	19.2	Collector - PAX												
		Delivery Form		L		Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Owner	64	Potassium Amyl Xanthate	A
		Delivery Size per Trailer				24,000	24,000	24,000	24,000	5	Vendor	112		1
		Number of Trailers per delivery	Maximum			2	2	2	2	1	Owner	64		A
		Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
		Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	16		A
		Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64		B
		Solution Mix Strength		% w/v		20	20	20	20	1	Owner	64		A
Supply Solution SG			t/m³		1.095	1.095	1.095	1.095	1	Owner	23		A	
Number of Storage Tanks					1	1	1	1	1	Owner	16		A	
Live Volume per Tank			m³		270	270	270	270	6	Calculated			B	
Residence Time		Actual	d		7	7	7	6	2	EPC		Residence time calculated using nominal consumption rates.	A	
		Required	d		7	7	7	7	1	Owner	10, 16		A	
Consumption														
Rougher Feed		Nominal	g/t fresh mill feed		40	40	40	40	1	Owner	16, 75		A	
Scavenger Feed		Nominal	g/t fresh mill feed		35	35	35	35	1	Owner	16, 75		A	
Cleaner Conditioning		Nominal	g/t fresh mill feed		16	16	16	16	1	Owner	16, 75		A	
Cleaner Scavenger		Nominal	g/t fresh mill feed		16	16	16	16	1	Owner	16, 75		A	
Rougher Feed		Nominal	L/h		629	592	629	699	6	Calculated				
Scavenger Feed		Nominal	L/h		551	518	551	611	6	Calculated				
Cleaner Conditioning		Nominal	L/h		252	237	252	280	6	Calculated				
Cleaner Scavenger		Nominal	L/h		252	237	252	280	6	Calculated				
Total Consumption		Nominal	L/h		1,683	1,585	1,683	1,869	6	Calculated				
		Nominal	m³/h		1.68	1.58	1.68	1.87	6	Calculated				
Consumption														
Rougher Feed		Design	g/t fresh mill feed		80	80	80	80	6	Calculated		100% Design Factor for pumping		
Scavenger Feed		Design	g/t fresh mill feed		70	70	70	70	6</					

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.3	Frother - Luprofroth 168										
	Delivery Form			Bulk Liquid	Bulk Liquid	Bulk Liquid		1	Owner	64	A
	Delivery Size per Trailer			26,000	26,000	26,000		5	Vendor		0
	Number of Trailers per delivery	Maximum		1	1	1		1	Owner	16	A
	Delivery Size	L		26,000	26,000	26,000		6	Calculated		
	Delivery System			Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	1	Owner	64	A
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Owner	64	B
	Solution Mix Strength	% w/w		100	100	100	100	1	Owner	64	A
	Supply Solution SG	tm³		1.082	1.082	1.082	1.082	1	Owner	16	A
	Number of Storage Tanks			1	1	1	1	2	EPC		A
	Live Volume per Tank	m³		60	60	60	60	1	Owner	16	A
	Residence Time	Minimum		7	7	7	7	1	Owner	10	A
		Actual		29	30	29	26	6	Calculated		
										Residence time calculated using nominal consumption rates.	
	Consumption										
	Rougher Feed	Nominal	g/t fresh mill feed	12	12	12	12	1	Owner	16, 75	A
	Scavenger Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	16, 75	A
	Cleaner Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	44, 75	A
	Cleaner Scavenger	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	16, 75	A
	Rougher Feed	Nominal	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Scavenger Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Scavenger	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Total Consumption	Nominal	L/h	87.2	82.1	87.2	96.9	6	Calculated		
			m³/h	0.09	0.08	0.09	0.10	6	Calculated		
	Consumption										
	Rougher Feed	Design	g/t fresh mill feed	24	24	24	24	6	Calculated		100% Design Factor for pumping
	Scavenger Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Cleaner Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Cleaner Scavenger	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Rougher Feed	Design	L/h	69.8	65.7	69.8	77.5	6	Calculated		100% Design Factor for pumping
	Scavenger Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Cleaner Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Cleaner Scavenger	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Total Consumption	Design	L/h	174.4	164.3	174.4	193.8	6	Calculated		100% Design Factor for pumping
			m³/h	0.17	0.16	0.17	0.19	6	Calculated		
19.4	Flocculant										
	Packaging Type			Bulk Powder	Bulk Powder	Bulk Powder	Bulk Powder	1	Owner	64	A
	Delivery Size per Trailer			22,500	22,500	22,500	22,500	1	Owner	64	A
	Number of Trailers per delivery	Maximum		1	1	1	1	1	Owner	64	A
	Delivery Size	kg		22,500	22,500	22,500	22,500	6	Calculated		
	Delivery System			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64	C
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Owner	64	B
	Storage Volume	Minimum	# deliveries	2	2	2	2	1	Owner	23	A
	Number of Mixing Tanks			1	1	1	1	1	Owner	16	A
	Mixing Tank Live Volume		m³	40	40	40	40	5	Vendor	36	B
	Solution Mix Strength	% w/v		0.30	0.30	0.30	0.30	1	Owner	44	A
	Mixes per Day per Tank			14	14	15	16	6	Calculated		Based on nominal Stage 2 consumption
	Flocculant Dilution Factor			10	10	10	10	1	Owner	103	B
	Number of Storage Tanks			1	1	1	1	5	Vendor	36	A
	Storage Tank Live Volume		m³	450	450	450	450	5	Vendor	36	A
	Residence Time		h	19	19	18	17	6	Calculated		Vendor advice. Based on nominal Stage 2 consumption
	Consumption										
	Flotation Tailings Thickener	Nominal	g/t	20	20	20	20	1	Owner	23	A
		Nominal	kg/h	61.4	57.2	60.8	68.2	6	Calculated		
		Nominal	L/h	20,473	20,054	22,738	22,738	6	Calculated		
		Design	L/h	40,946	38,108	40,557	45,476	6	Calculated		
	Concentrate Pre-Leach Thickener	Nominal	g/t	70	70	70	70	1	Owner	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,949	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Concentrate Thickener	Nominal	g/t	70	70	70	70	1	Owner	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,948	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Total Nominal Consumption	Stage 2	g/h	71,909	71,742	75,417	79,908	6	Calculated		
		Stage 2	L/h	23,970	23,914	25,139	26,636	6	Calculated		
		Stage 2	m³/h	24.0	23.9	25.1	26.6	6	Calculated		
		Stage 2	t/a	588	587	617	654	6	Calculated		
	Total Design Consumption	Stage 2	g/h	95,000	95,000	95,000	95,000	2	EPC		Based on 28.35Mtpa Total Consumption for Stage 2 when the Regrind Mill was still included. Regrind Mill was moved post tender phase.
		Stage 2	t/a	777	777	777	777	6	Calculated		
		Stage 2	m³/h	31.7	31.7	31.7	31.7	6	Calculated		Neat Concentration
		Stage 2	L/h	47,938	47,826	50,277	53,270	6	Calculated		100% Design Factor
		Stage 2	m³/h	47.9	47.8	50.3	53.3	6	Calculated		100% Design Factor
19.6	Silo Capacity	Actual	t	45	45	45	45	5	Vendor	36	A
			d	26	26	25	23	6	Calculated		Based on nominal Stage 2 consumption
	pH Modifier - Quicklime										
	Type			New Powder	New Powder	New Powder	New Powder				
	Packaging Type			Bulk	Bulk	Bulk	Bulk	1	Owner	64	B
	Delivery Size per Trailer		kg	30,000	30,000	30,000	30,000	1	Owner	64	A
	Number of Trailers Per Delivery	#		3	3	3	3	1	Owner	64	A
	Delivery Size	kg		90,000	90,000	90,000	90,000	6	Calculated		
	Preferred Delivery Method			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64	C
	Bulk Density		tm³	0.95	0.95	0.95	0.95	1	Owner	16	A
	Solids SG			3.37	3.37	3.37	3.37	1	Owner	16	A
	Water SG			1.07	1.07	1.07	1.07	1	Owner	16	A
	Solution Mix Strength	% w/w		20	20	20	20	1	Owner	16	A
	Lime Slurry Density	tm³		1.24	1.24	1.24	1.24	6	Calculated		Site metallurgical accounting assumption Cyclone overflow density
	Feed Size										
	F100		mm	6.7	6.7	6.7	6.7	1	Owner	116	1
	F80		µm	1,150	1,150	1,150	1,150	1	Owner	116	1
	Product Size		µm	38	38	38	38	1	Owner	116	1
	Dry Consumption										
	Concentrate CIL - Nominal	concentrate basis	kg/t	51.2	51.2	51.2	51.2	1	Owner	23	A
	Concentrate CIL - Design	concentrate basis	kg/t	120.1	120.1	120.1	120.1	1	Owner	16	A
	Flotation Tailings CIL - Nominal	tailings CIL basis	kg/t	3.20	3.20	3.20	3.20	1	Owner	23	A
	Flotation Tailings CIL - Design	tailings CIL basis	kg/t	4.50	4.50	4.50	4.50	1	Owner	16	A
	Total Dry Consumption										
	Nominal	@ 93.4% Availability	th	13.9	14.8	15.4	15.5	6	Calculated		
	Nominal	@ 75% Lime Utilisation	th	17.3	18.5	19.2	19.3	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033
	Design	@ 93.4% Availability	th	23.2	25.9	26.7	25.8	6	Calculated		
	Design	@ 75% Lime Utilisation	th	28.9	32.2	33.2	32.1	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033
	Milk of Lime Consumption										
	Nominal		m³/h	56.2	59.9	62.2	62.4	6	Calculated		
	Peak		m³/h	93.5	104.4	107.7	104.0	6	Calculated		
	New Circuit										
	Storage Silo Capacity	per Silo	t	1,132	1,132	1,132	1,132	5	Vendor	118	1
	Number of Silos			2	2	2	2	5	Vendor	118	1
	Residence Time	Actual	d	7	6	6	6	6	Calculated		Based on nominal consumption
	Lime Slaker Type			MIII	MIII	MIII	MIII	1	Owner	43	A
	Number of Slakers			2	2	2	2	5	Vendor	118	1
	Slaker Power		kW/mill	6 x 55	6 x 55	6 x 55	6 x 55	5	Vendor	118	2
	Utilisation	%		75	75	75	75	1	Owner	116	A
	Mill Throughput	Design	thr/mill	17	17	17	17	5	Vendor	118	2
			t/hr	33	33	33	33	5	Vendor	118	2
	Make-up Water Type			Raw Water	Raw Water	Raw Water	Raw Water	1	Owner	103	A
	Milk of Lime Storage Tank Live Volume	Max	m³	524	524	524	524	1	Owner	107	0
	Number of Storage Tanks			1	1	1	1	1	Owner		A
	Residence Time	Actual	h	9.3	8.8	8.4	8.4	6	Calculated		
	Ringmain Flow Factor			2.5	2.5	2.5	2.5	2	EPC		Applied to Peak flow consumption rates
	Lime Ringmain Pumps	Required	m³/h	234	261	269	260	6	Calculated		C
		Selected	m³/h	300	300	300	300	2	EPC		C

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.7	Hydrogen Peroxide										
	Type			Liquid	Liquid	Liquid	Liquid	1	Owner	64	A
	Supply Strength	% w/v		70	70	70	70	1	Owner	64	B
	Supply Solution SG	tim³		1.29	1.29	1.29	1.29	5	Vendor	111	1
	pH			4.3	4.3	4.3	4.3	1	Owner	44	B
	Delivery Form			Bulk, Isolainer	Bulk, Isolainer	Bulk, Isolainer	Bulk, Isolainer	1	Owner	64	B
	Delivery Size per Trailer	L		24,000	24,000	24,000	24,000	5	Vendor	64	0
	Number of Trailers Per Delivery	#		2	2	2	2	1	Owner	64	
	Delivery Size	L		48,000	48,000	48,000	48,000	6	Calculated		B
	Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64	B
	Delivery Side			Central Top	Central Top	Central Top	Central Top	1	Owner	64	B
	No. Dosing Points			3	3	3	3	1	Owner		
	Consumption Rate - Nominal	each Inline Mixer	L/h	140	140	140	140	6	Owner	35	One dosing pump per mixer.
		Total	L/h	420	420	420	420	6	Calculated		A
	Consumption Rate - Design	each Inline Mixer	L/h	280	280	280	280	6	Calculated		
		Total	L/h	840	840	840	840	6	Calculated		
	Number of Storage Tanks			1	1	1	1	2	EPC	316 SS Storage Tank	A
	Storage Tank Live Volume	Per Tank	m³	84.35	84.35	84.35	84.35	2	EPC	2000-F72-TQY-ME-10322_1	1
		Total	m³	84	84	84	84	6	Calculated		
	Hydrogen Peroxide Storage Tank Residence Time			7	7	7	7	1	Owner	10	
	Required Residence Time	Nominal	days	8	8	8	8	6	Calculated		B
	Nominal Usage Rate		days								
19.8	Sodium Cyanide										
	Delivery Form			New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	1	Owner	64	B
	Delivery Size per Trailer	L		22,000	22,000	22,000	22,000	1	Owner	64	A
	Number of Trailers per delivery	Maximum		2	2	2	2	1	Owner	64	A
	Delivery Size	L		44,000	44,000	44,000	44,000	6	Calculated		
	Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64	C
	Delivery Side			Central Top	Central Top	Central Top	Central Top	1	Owner	64, 83	B
	Solution Mix Strength	% w/v		29	29	29	29	1	Owner	64	4
	Supply Solution SG	tim³		1.24	1.24	1.24	1.24	2	EPC		4
	Consumption										
	Flotation Concentrate CIL	Operating Range	kg/t CIL Feed	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	1	Owner	23, 44	Dosed to six tanks
		No. of Dose Points	#	6	6	6	6	1	Owner	23, 44, 103	Dosed to six tanks
		Nominal	kg/t CIL per tank	3.58	3.58	3.58	3.58	1	Owner	44	Dosed to six tanks. Total of 21.50 kg/t as per Owner provided data.
		Nominal	kg/t CIL Total	21.5	21.5	21.5	21.5	6	Calculated		
		Nominal	kg/d	38,658	63,733	63,740	43,093	6	Calculated		
	Flotation Concentrate CIL	Design	kg/t CIL per tank	10	10	10	10	1	Owner	23, 44	Dosed to six tanks
		Design	kg/t CIL Total	60	60	60	60	6	Calculated		A
		Design	kg/d	107,882	149,951	149,973	120,260	6	Calculated		of NaCN
	Flotation Tailings CIL	Operating Range	kg/t CIL Feed	0.04-0.24	0.04-0.24	0.04-0.24	0.04-0.24	1	Owner	23	
		No. of Dose Points	#	4	4	4	4	1	Owner	103	CIL4 Feed Distribution Box, Leach tank 1, 2 and Adsorption Tank 1
	Flotation Tailings CIL	Nominal	kg/t CIL Total	0.22	0.22	0.22	0.22	1	Owner	23	Dosing split between dosing points
		Nominal	kg/d	16,637	16,678	16,648	18,480	6	Calculated		
	Flotation Tailings CIL	Design	kg/t CIL Feed	0.33	0.33	0.33	0.33	1	Owner	16	Dosing split between dosing points
		Design	kg/d	24,956	23,517	24,972	27,720	6	Calculated		
	Elution Circuit 1	CIL2/3	m³/strip	1.2	1.2	1.2	1.2	1	Owner	16	
		Min. instantaneous	m³/h	2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 2	CIL2/3	m³/strip	1.2	1.2	1.2	1.2	1	Owner	16	
		Min. instantaneous	m³/h	2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 3	CIL4	m³/strip	1.51	1.51	1.51	1.51	6	Calculated		
		Min. instantaneous	m³/h	3.03	3.03	3.03	3.03	6	Calculated		
	Elution Consumption	Nominal	kg/d	2,375	2,375	2,375	2,375	6	Calculated		
		Design	kg/d	4,750	4,750	4,750	4,750	6	Calculated		100% Design Factor
	Intensive Leach		kg/batch	230	230	230	230	5	Vendor	71	Based on CS10000 Acacia unit
		Nominal	kg/d	230	230	230	230	6	Calculated		
		Design	kg/d	460	460	460	460	6	Calculated		100% Design Factor
	Nominal Consumption		kg/d	57,900	72,016	72,993	64,178	6	Calculated		
			kg/h	2,413	3,001	3,041	2,674	6	Calculated		
		NaCN solution (29%w/w)	L/h	6,734	8,375	8,489	7,464	6	Calculated		
			kg/t plant feed	0.77	1.01	0.97	0.77	6	Calculated		
	Design Consumption		kg/d	138,048	178,678	180,155	153,190	6	Calculated		
			kg/h	5,752	7,445	7,506	6,383	6	Calculated		
		NaCN solution (29%w/w)	L/h	16,054	20,779	20,951	17,815	6	Calculated		
			kg/t plant feed	1.83	2.51	2.39	1.83	6	Calculated		
	New Circuit										
	Number of Storage Tanks			2	2	2	2	1	Owner	83	
	Live Volume per Tank	Each Tank	m³	1,250	1,250	1,250	1,250	1	Owner	83	
	Residence Time	Actual	d	15.5	12.4	12.3	14.0	6	Calculated		Based on nominal consumption.
	Ringmain Flow Factor			2.0	2.0	2.0	2.0	2	EPC		Applied to Design flow consumption rates
	Cyanide Ringmain Pumps	Required	m³/h	32.1	41.6	41.9	35.6	6	Calculated		
		Selected	m³/h	50	50	50	50	2	EPC		
19.9	Oxygen			New						Owner Supplied Package	
	Flotation Tails CIL										
	Consumption Rate	Nominal	Nm³/t mill feed		0.17			6	Calculated	91	
	Flotation Concentrate CIL										
	Consumption Rate	Nominal	Nm³/h conc		13			1	Owner	23	
	Total Consumption										
		Nominal	t/d		53.2			6	Calculated	91	
		Maximum	t/d		70.8			6	Calculated	91	
	New Oxygen Plant										
	VPSA Capacity	Selected	t/d		70			1	Owner	47	Owner Supplied Package
19.10	Lead Nitrate										
	Delivery Form			New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	1	Owner	64	A
	Delivery Size per Trailer	L		24,000	24,000	24,000	24,000	5	Vendor	112	1
	Number of Trailers per delivery	Maximum		2	2	2	2	1	Owner	64	A
	Delivery Size	L		48,000	48,000	48,000	48,000	6	Calculated		
	Preferred Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Owner	64	A
	Delivery Side			Central Rear	Central Rear	Central Rear	Central Rear	1	Owner	64	B
	Solution Mix Strength	% w/v		40	40	40	40	1	Owner	64	A
	Supply Solution SG	tim³		1.17	1.17	1.17	1.17	1	Owner	16	A
	Consumption										
19.11	Sodium Hydroxide (Caustic)										
	Delivery Form			New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	1	Owner	64	A
	Delivery Size per Trailer	L		20,000	20,000	20,000	20,000	1	Owner	64	A
	Number of Trailers per delivery	Maximum		3	3	3	3	1	Owner	64	A
	Delivery Size	L		60,000	60,000	60,000	60,000	6	Calculated		
	Preferred Delivery System			Pumped Transfer From Truck	Pumped Transfer From Truck	Pumped Transfer From Truck	Pumped Transfer From Truck	1	Owner	64	B
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Owner	64	B
	Solution Mix Strength	% w/v		50	50	50	50	1	Owner	64	A
	Supply Solution SG	tim³		1.50	1.50	1.50	1.50	1	Owner	16	A
	Consumption										
	Elution Circuit 1		m³/strip		1.7			1	Owner	16	Existing
		Min. instantaneous	m³/h		3.4			6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 2		m³/strip		1.7			1	Owner	16	
		Min. instantaneous	m³/h		3.4			6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 3		m³/strip		0.85			6	Calculated		
		Min. instantaneous	m³/h		1.70			6	Calculated		
	Intensive Leach		kg/batch		29			5	Vendor	71	Based on CS10000 Acacia unit
			L/batch		39			6	Calculated		
	Consumption		kg/d		1,913			6	Calculated		
	Elution Circuit 1		kg/d		1,913			6	Calculated		
	Elution Circuit 2		kg/d		1,277			6	Calculated		
	Elution Circuit 3		kg/d		58			6	Calculated		Based on two batches per day
	Intensive Leach		kg/d								
	Calculated Consumption	Nominal	kg/d		5,160			6	Calculated		
		Nominal	kg/h		215			6	Calculated		
		Nominal	L/h		287			6	Calculated		
	New Circuit										
	Number of Storage Tanks			1	1	1	1	2	EPC		C
	Live Volume		m³	250	250	250	250	2	EPC		A
	Residence Time	Nominal	d	36	36	36	36	6	Calculated		Based on Nominal Consumption
	Ringmain Flow Factor			2.5	2.5	2.5	2.5	2	EPC		Applied to combined instantaneous flowrate from Elution Circuits 1 & 2
	Caustic Ringmain Pumps	Required	m³/h	17	17	17	17	6	Calculated		
		Selected	m³/h	20	20	20	20	2	EPC		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.12	Hydrochloric Acid			New	Bulk Liquid						
	Delivery Form			20,000			1	Owner	64		A
	Delivery Size per Trailer			3			1	Owner	64		A
	Number of Trailers per delivery	Maximum		60,000			1	Owner	64		A
	Delivery Size	L		Pumped Transfer From Truck			6	Calculated			A
	Preferred Delivery System			Passenger			1	Owner	64		A
	Delivery Side			32			1	Owner	64		B
	Supply Strength	% w/w		1.16			1	Owner	64		A
	Supply Solution SG	t/m³					1	Owner	16		A
	Consumption										
	Acid Wash Column 1	m³/strip		2.7			1	Owner	16		A
		Min. instantaneous		9.5			6	Calculated			
	Acid Wash Column 2	m³/strip		2.7			1	Owner	16		A
		Min. instantaneous		9.5			6	Calculated			
	Acid Wash Column CIL4	m³/strip		1.5			6	Calculated			
		Min. instantaneous		5.2			6	Calculated			
	Consumption										
	Acid Wash Column 1	kg/d		1,503			6	Calculated			
	Acid Wash Column 2	kg/d		1,503			6	Calculated			
	Acid Wash Column CIL4	kg/d		1,085			6	Calculated			
	Calculated Consumption										
		kg/d		4,092			6	Calculated			
		kg/h		170			6	Calculated			
		L/h		256			6	Calculated			
		L/h		459			6	Calculated			
	New Circuit										
	Number of Storage Tanks			1			1	Owner	24		A
	Live Volume per Tank	m³		200			2	EPC			A
	Residence Time	Nominal	d	18			6	Calculated		Based on Nominal Consumption	
	Dilute Acid Storage Tank Live Volume	m³		25			2	EPC	110		0
	Neat Acid Dosing Pumps	Required	m³/h	20			2	EPC		2 off Elution Circuit 1/2 strips occurring simultaneously	C
	Dilute Acid Dosing Pump	Minimum	m³/h	64			2	EPC			0
19.13	Grinding Media										
	SAG Mill										
	Number of Bunkers			2			1	Owner	16		A
	Bunker Storage Size	m³		35			1	Owner	16		A
	Media Type			Forged Steel			1	Owner	16, 38		A
	Media Diameter	mm		125			1	Owner	16, 38		A
	Bulk Density	t/m³		4.65			1	Owner	16		A
	Package Form			Bulk			1	Owner	16		A
	Delivery Size	t		25			1	Owner	23		A
	Trailers per Delivery	#		1			1	Owner	23		A
	Consumption Rate - Existing SAG	kg/t mill feed		0.35			1	Owner	16, 38		A
	Consumption Rate - New SAG	kg/t mill feed		0.40			1	Owner	16, 38		A
	Total Consumption	t/a		9,759			6	Calculated			
	Storage Capacity	t		325.5			6	Calculated			
	Residence Time	d		12			6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets	1.5			1	Owner	16	CAT972 Bucket	A
	Ball Mill										
	Number of Bunkers			1			1	Owner	16		A
	Bunker Storage Size	m³		35			1	Owner	16		A
	Media Type			High Chrome			1	Owner	16, 38		A
	Media Diameter	mm		75			1	Owner	16, 38		A
	Bulk Density	t/m³		4.75			1	Owner	16		A
	Package Form			Bulk			1	Owner	16		A
	Delivery Size	t		25			1	Owner	23		A
	Trailers per Delivery	#		1			1	Owner	23		A
	Consumption Rate	kg/t mill feed		0.20			1	Owner	16, 38		A
	Consumption	t/a		5,148			6	Calculated			
	Storage Capacity	t		166.3			6	Calculated			
	Residence Time	d		12			6	Calculated			
	Indexer/Charger Storage Capacity	Minimum	Buckets	1.5			1	Owner	16	CAT972 Bucket	A
	UFG Mill										
	Media Type			Ceramic			1	Owner	16		A
	Media Diameter	mm		2.5			1	Owner	16		A
	Bulk Density	t/m³		2.2			1	Owner	16		A
	Package Form			Bulk Bag			1	Owner	16		A
	Package Weight	kg		2,000			1	Owner	16		A
	Addition Rate	kg/t mill feed		0.56			1	Owner	10		B
		UFG Mill 1		0.0			6	Calculated			
		UFG Mill 1		0.0			6	Calculated			
		UFG Mill 2		21.0			6	Calculated			
		UFG Mill 2		503.4			6	Calculated			
		UFG Mill 3		21.0			6	Calculated			
		UFG Mill 3		503.4			6	Calculated			
	Site Storage Capacity	Minimum	# Bags	50			1	Owner			
		Design	# Bags	54			2	EPC			
		days		107			6	Calculated		Navisworks model. Designed to achieve the minimum. UFG Mill 1, 2 & 3 Consumption	B
	Lime Slaking Mills										
	Media Type			Steel			5	Vendor	118		2
	Media Diameter	mm		25.0			5	Vendor	118		2
	Bulk Density	t/m³		4.85			5	Vendor	118		2
	Total Consumption	kg/mill/week		170			5	Vendor	118	Estimated. Dependant on lime slaking usage and hardness	2
		kg/week total		340			6	Calculated		Estimated. Dependant on lime slaking usage and hardness	
		kg/t		10.2			6	Calculated		Estimated using vendor provided mill ball consumption and lime design throughput	
	Package Form			Bulk Bag			1	Owner	118		2
	Media Hopper Capacity	UFG Mill 2	m³	14			5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 2	days	61.2			6	Calculated			
	Residence Time	# Bags		15.4			6	Calculated			
	Media Hopper Capacity	UFG Mill 3	m³	14			5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 3	days	61.2			6	Calculated			
	Residence Time	# Bags		15.4			6	Calculated			
19.14	Antiscalant										
	Type			FF585			1	Owner	16	Dosing pump for each dosing point	A
	Addition Rate - Gravity Concentrators	ml/min		40			1	Owner	16	New addition in Stage 1, to be dosed into gravity concentrator fluidisation water line	A
	Addition Rate - Kiln Scrubber (Existing)	ml/min		45			1	Owner	16	New addition in Stage 2	A
	Addition Rate - CIL4 Kiln Mercury Abatement System	ml/min		61			1	Owner	44	New addition in Stage 2	B
	Total Consumption	ml/min		146			6	Calculated			
	Number of IBC's for Storage	No.		2			6	Calculated		Based on 7 days storage & 1,000 L IBC volumes	
	Type			FF540			1	Owner	16	Dosing pump for each dosing point	A
	Addition Rate - Elution Circuit 1 (Existing)	ml/min		56			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 2 (Existing)	ml/min		70			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 1 Heat Exchanger (Existing)	ml/min		24			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 2 Heat Exchanger (Existing)	ml/min		30			1	Owner	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit CIL4 (New)	ml/min		95			1	Owner	44	New addition in Stage 2	B
	Addition Rate - Elution Circuit CIL4 Heat Exchanger (New)	ml/min		41			1	Owner	44	New addition in Stage 2	B
	Total Consumption	ml/min		316			6	Calculated			
	Number of IBC's for Storage	No.		4			6	Calculated		Based on 7 days storage & 1,000 L IBC volumes	
19.15	LPG										
	Type			Propane			1	Owner	16		A
	Elution Burner 1 Range	High Flame	MJ/h	4,900			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame	MJ/h	2,000			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Elution Burner 2 Range	High Flame	MJ/h	4,900			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame	MJ/h	2,000			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	CIL4 Elution Burner Range	Pre-Heat	L/h	823			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
		Steady State	L/h	708			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 1		MJ/h	4,500			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 2		MJ/h	4,500			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 3		MJ/h	4,500			1	Owner	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regenerative Thermal Oxidiser		MJ/h	1,580			1	Owner	88	Existing equipment fed from upgraded LPG Storage Vessel	B
	Smelting Furnace	Maximum	MJ/h	2,000			1	Owner	16	Existing equipment fed from dedicated Goldroom LPG Storage Vessels	A
	Consumption Rate	Plant Feed Basis	L/t	0.40			6	Calculated	50	Inclusive of Regeneration Kiln 3 and RTO but not the Goldroom Smelting Furnace. Calculated from a 25.5 Mtpa basis.	B
	Number of LPG Storage Vessels (Upgraded)		kL	1			1	Owner	88	Client supply	B
	Vessel Capacity			161			1	Owner	88	Client supply	B
	Residence Time	Actual	d	5.3			6	Calculated		As per source 82, Owner confirm and accept they won't achieve the 7 days	

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
				1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
20.0	WATER SERVICES - EXISTING											
20.1	Saline Bore Water											
	Source						1	Owner	16		A	
	Storage Method 1						1	Owner	16		A	
	Live Volume	m³					1	Owner	16		A	
	Dam Water Pumps	#					1	Owner	23		A	
	Pump Model						1	Owner	23		A	
	Storage Method 2						1	Owner	16		A	
	Live Volume	m³					1	Owner	16		A	
	Dam Water Pumps	#					1	Owner	106		0	
	Pump Model						1	Owner	23		A	
Total Live Volume	m³					6	Calculated					
20.2	Mill Circuit Water											
	Source			Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F	Concentrate Pre-Leach Thickener O/F	Concentrate Pre-Leach Thickener O/F	1	Owner		B	
	Storage Method			Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	1	Owner	16	A	
	Live Volume	m³		1,500	1,500	1,500	1,500	1	Owner	16	A	
	Residence Time	min		100	94	94	100	6	Calculated			
	Water Demand - Nominal	m³/h		896	958	958	899	6	Calculated			
	Water Demand - Design	m³/h		1,800	1,800	1,800	1,800	1	Owner	16	A	
	Design Flowrate per Pump	m³/h		1,800	1,800	1,800	1,800	1	Owner	16	A	
	Pump Configuration			Duty / Standby APPS1-300	Duty / Standby APPS1-300	Duty / Standby APPS1-300	Duty / Standby APPS1-300	1	Owner	16	A	
	Pump Model							1	Owner	23	A	
20.3	Cyanide Water											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Live Volume	m³					1	Owner	16		A	
	Cyanide Water Pump Configuration						1	Owner	16		A	
	Cyanide Water Pump Model						1	Owner	23		A	
	Cyanide Transfer Pump Configuration						1	Owner	16		A	
	Cyanide Transfer Pump Model						1	Owner	23		A	
	Design Flowrate per Pump	m³/h					1	Owner	16		A	
	Pump Configuration						1	Owner	16		A	
20.4	Scheme Water											
	Water Demand	m³/h		172	172	173	175	6	Calculated			
	Scheme Water Quality											
	SG	-		1.00	1.00	1.00	1.00	1	Owner	1	B	
	TDS	mg/L		454	454	454	454	1	Owner	1	B	
	pH			8.0	8.0	8.0	8.0	1	Owner	1	B	
	Chlorides	mg/L		232	232	232	232	1	Owner	1	B	
	Fire Water / Safety Shower / Potable Water						1	Owner	69	By Others		B
	Gland Water											
	Source						1	Owner	16		A	
Storage Method						1	Owner	16		A		
Live Volume	m³					1	Owner	16		A		
Pump Configuration						1	Owner	16		A		
Pump Model						1	Owner	23		A		
20.7	Eastern Borefield Transfer Dam											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	79		B	
	Live Volume	m³					1	Owner	16		A	
	Pump Configuration						1	Owner	16		A	
	Saline Gravity Dam											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	79		B	
Live Volume	m³					1	Owner	16		A		
20.8	Saline Gravity Dam											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	79		B	
	Live Volume	m³					1	Owner	16		A	
	Katbais Saline Dam											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	79		B	
	Live Volume	m³					1	Owner	16		A	
Pump Configuration						1	Owner	16		A		
21.0	Saline Transfer Dam											
	Source						1	Owner	16		A	
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	79		B	
	Live Volume	m³					1	Owner	16		A	
	Pump Configuration						1	Owner	16		A	
	Southern Surge Tank											
	Storage Method						1	Owner	16		A	
	Destination						1	Owner	16		A	
	Capacity	m³					1	Owner	16		A	
21.0	Saline Transfer Dam											
21.1	Process Water											
	Source						1	Owner	10		A	
	Storage Method						9	Agreed	10		A	
	Live Volume	m³					9	Agreed	10		A	
	Residence Time	Minimum min					1	Owner	16		A	
	Water Demand	Calculated m³/h					6	Calculated				
	Pump Configuration						9	Agreed	10		A	
	Process Water Quality											
	SG	-					1	Owner	1		B	
	TDS	mg/L					1	Owner	1		B	
pH	-					1	Owner	1		B		
Chlorides	mg/L					1	Owner	1		B		
21.2	Raw Water											
	Source						1	Owner	43	Supplies the Raw Water Tank		A
	Storage Method						1	Owner	68		B	
	Configuration						1	Owner	68		B	
	Total Live Volume	Actual m²					1	Owner	68		B	
	Total Live Volume	Required m²					6	Calculated				
	Residence Time	Actual min					6	Calculated				
	Residence Time	Required min					9	Agreed	10, 16	Nominal Basis		B
	Total Nominal Water Demand (water consumption)	m³/h					2	EPC	68	Note: Pump design for instantaneous flows in accordance with mechanical calculation.		B
	Pump Configuration						1	Owner	16		A	
Source (Greenfields)						1	Owner	69	By Others		B	
Live Volume	m³					1	Owner	114	By Others		1	
21.3	Gland Water											
	Source						1	Owner	43		A	
	Gland Water Demand	Design m³/h					2	EPC			0	
	Low Pressure Gland Water Pump Configuration						1	Owner	43		A	
	High Pressure Gland Water Pump Configuration						1	Owner	43		A	
	22.0	AIR SERVICES										
	22.1	Low Pressure Flotation Air										
		Number of Flotation Blowers						1	Owner	49		A
		Operating Configuration						1	Owner	49		A
		Type						1	Owner	49		A
Capacity - Each		Design FAD m³/h					1	Owner	49		A	
Capacity - Total		Design FAD m³/h					6	Calculated				
Pressure		Design kPa(g)					1	Owner	49		A	
High Pressure Air												
Number of Compressors							1	Owner	43		A	
Operating Configuration							1	Owner	43		A	
Type						5	Vendor	45		B		
Capacity - Each	NTP Conditions L/s					5	Vendor	45		A		
Pressure	kPa(g)					5	Vendor	45		B		
Duty						2	EPC			A		